DRAWING LIST FOR TEST PACKAGE

AIR- SV-511

PROJECT NO.: 22330D

PROJECT TITLE : M15X Ph4 Project

No	PKG NO	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING NO	SHEET	REV	P&ID NO	REV	
1	511	5FA	25A-AIR-SV-ST1-N-9743-0	1/1	0	CJ-M15W13-PR-A20-000083-A	2	
2	511	5FA	25A-AIR-SV-ST1-N-9744-0	1/1	0	CJ-M15W13-PR-A20-000083-A	2	
3	511	5FA	25A-AIR-SV-ST1-N-9745-0	1/1	0	CJ-M15W13-PR-A20-000083-A	2	
4	511	5FA	40A-AIR-SV-ST1-N-9532-0	1/6	0	CJ-M15W14-PR-A20-000142	1A	
5	511	5FA	40A-AIR-SV-ST1-N-9532-0	2/6	0	CJ-M15W14-PR-A20-000142	1A	
6	511	5FA	40A-AIR-SV-ST1-N-9532-0	3/6	0	CJ-M15W14-PR-A20-000142	1A	
7	511	5FA	40A-AIR-SV-ST1-N-9532-0	4/6	0	CJ-M15W14-PR-A20-000142	1A	
8	511	5FA	40A-AIR-SV-ST1-N-9532-0	5/6	0	CJ-M15W14-PR-A20-000142	1A	
9	511	5FA	40A-AIR-SV-ST1-N-9532-0	6/6	0	CJ-M15W14-PR-A20-000142	1A	
10	511	5FA	40A-AIR-SV-ST1-N-9541-0	1/1	0	CJ-M15W14-PR-A20-000102	1A	
11	511	5FA	40A-AIR-SV-ST1-N-9542-0	1/1	0	CJ-M15W14-PR-A20-000103	1A	
12	511	5FA	40A-AIR-SV-ST1-N-9543-0	1/1	0	CJ-M15W14-PR-A20-000104	1A	
13	511	5FA	40A-AIR-SV-ST1-N-9544-0	1/1	0	CJ-M15W14-PR-A20-000105	1A	
14	511	5FA	40A-AIR-SV-ST1-N-9562-0	1/1	0	CJ-M15W14-PR-A20-000107	1A	
15	511	5FA	40A-AIR-SV-ST1-N-9563-0	1/1	0	CJ-M15W14-PR-A20-000108	1A	
16	511	5FA	40A-AIR-SV-ST1-N-9564-0	1/1	0	CJ-M15W14-PR-A20-000109	1A	
17	511	5FA	40A-AIR-SV-ST1-N-9565-0	1/1	0	CJ-M15W14-PR-A20-000110	1A	
18	511	5FA	65A-AIR-SV-ST1-N-9526-0	1/3	0	CJ-M15W13-PR-A20-000083-A	2	
19	511	5FA	65A-AIR-SV-ST1-N-9526-0	2/3	0	CJ-M15W13-PR-A20-000083-A	2	
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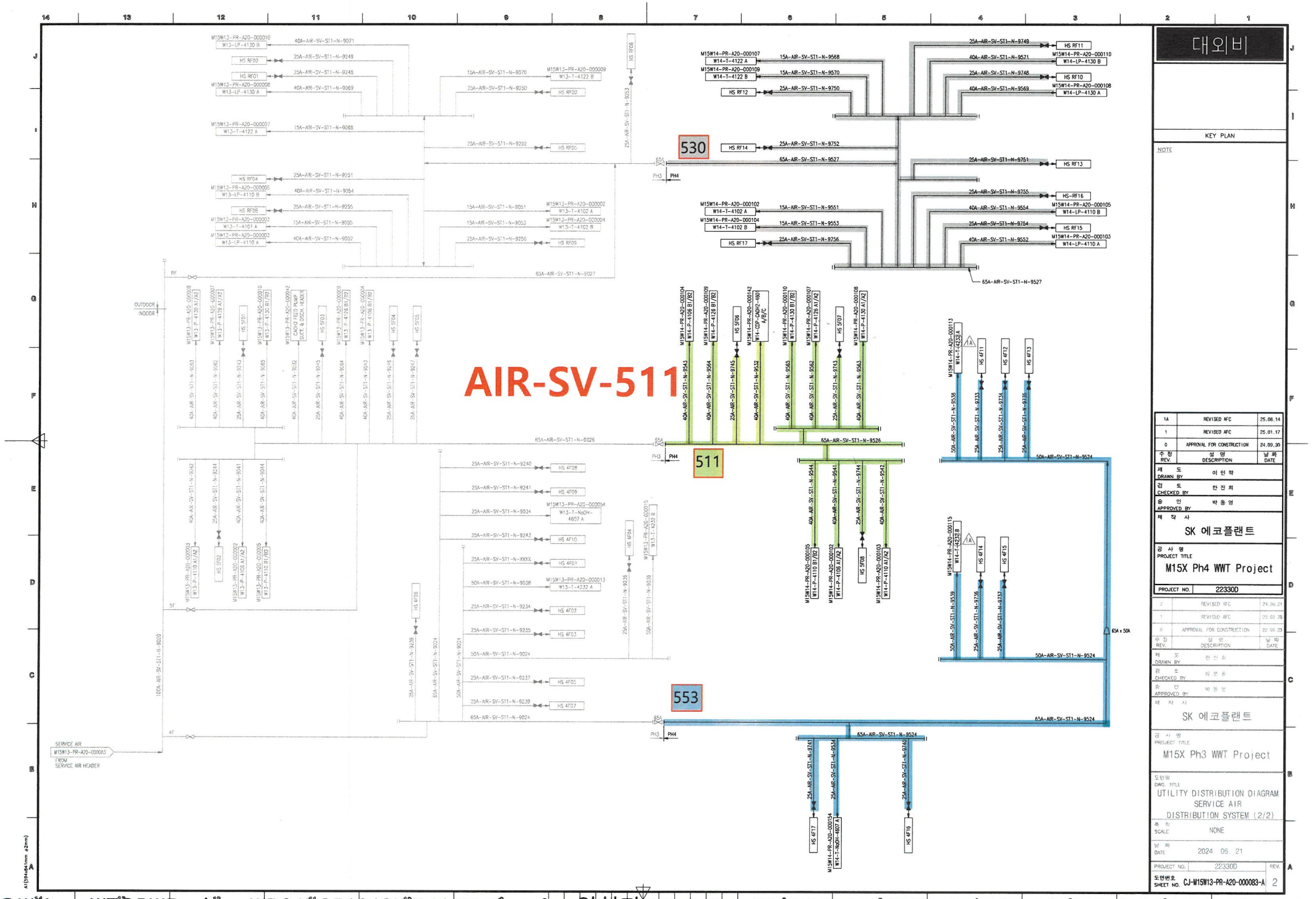

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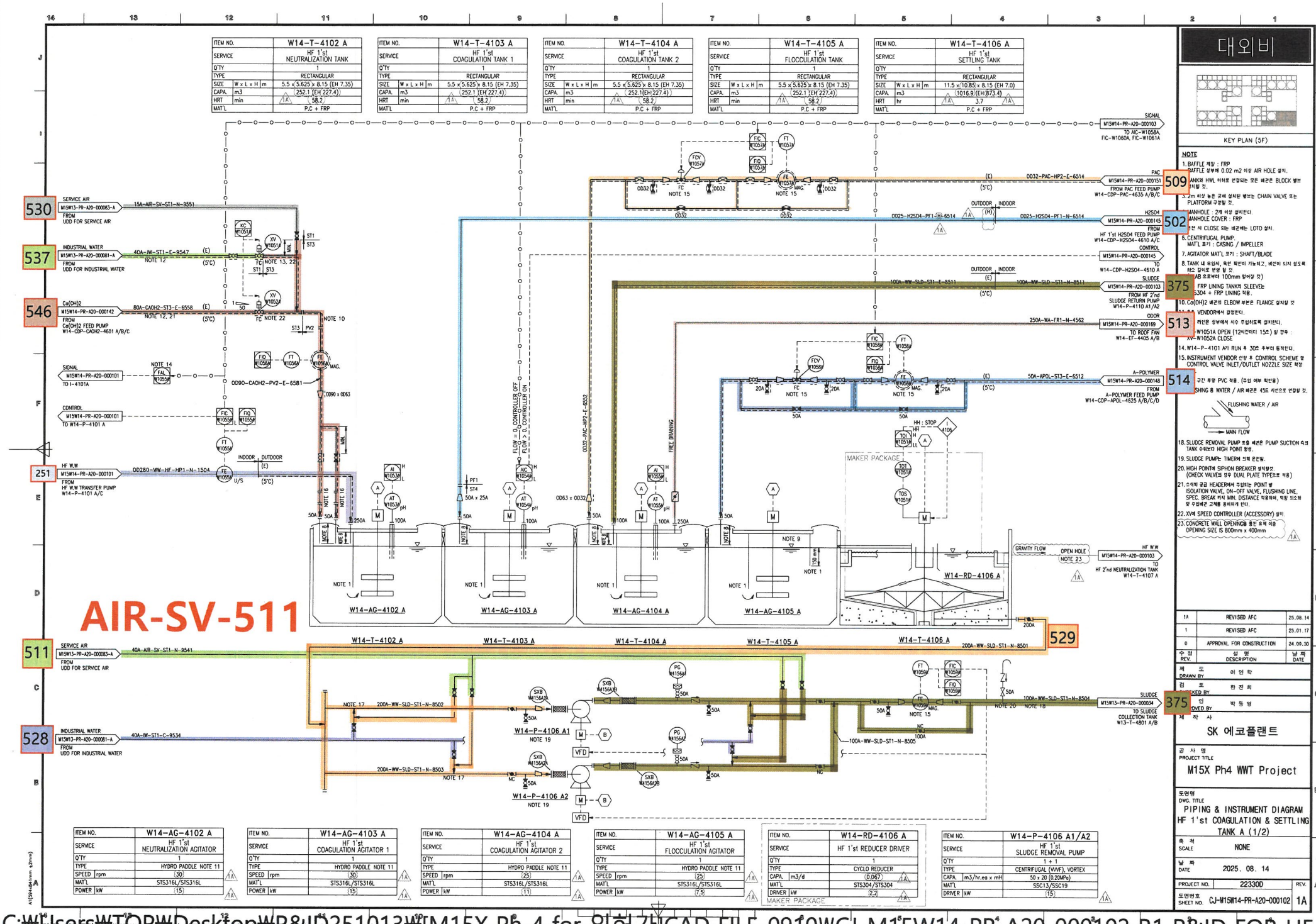

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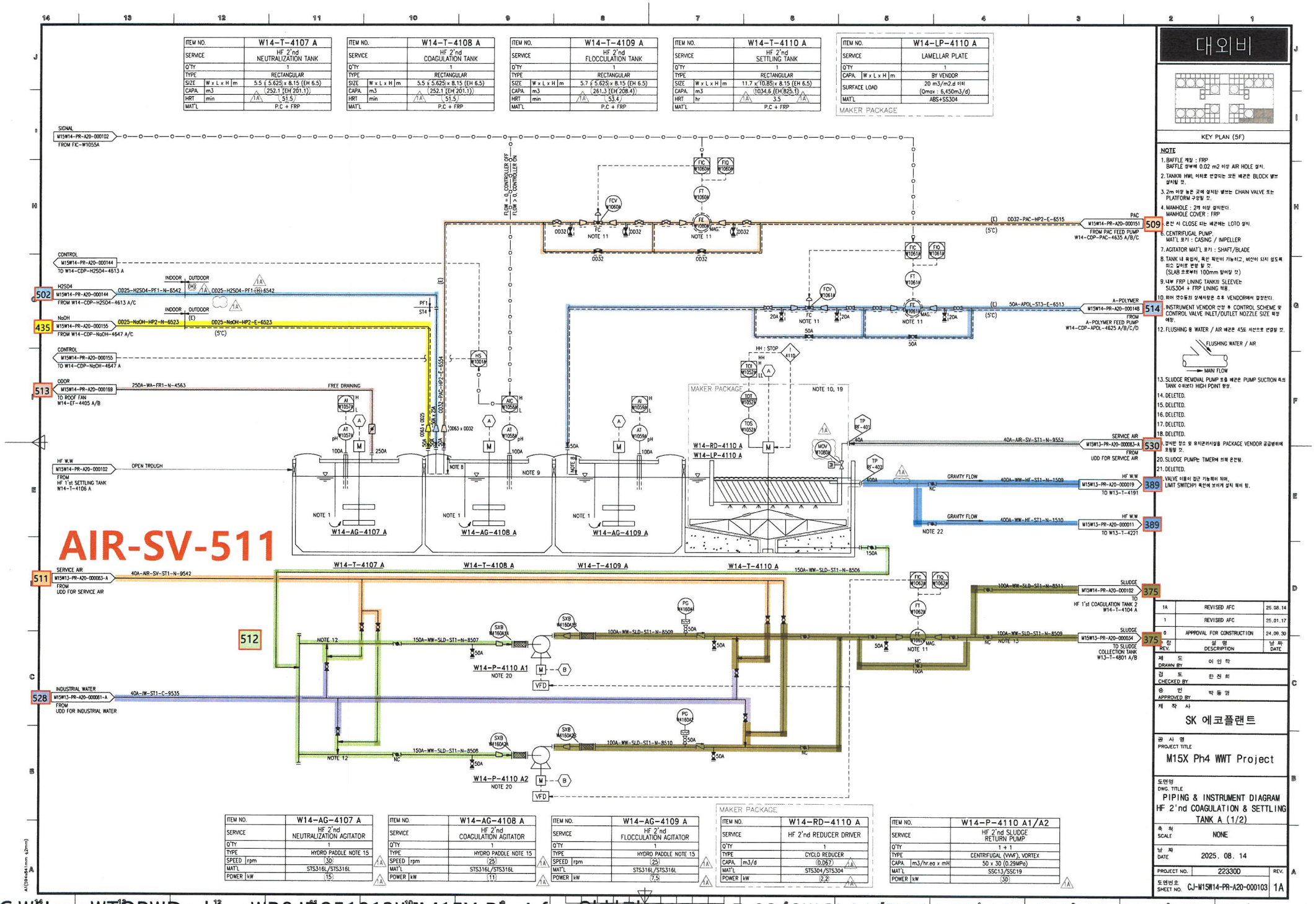

ITEM NO.	W14-T-4107 A
SERVICE	HF 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE W x L x H | m	5.5 x 5.625 x 8.15 (EH 6.5)
CAPA. m3	(252.1 (EH 201.1))
HRT min	(51.5)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4108 A
SERVICE	HF 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE W x L x H | m	5.5 x 5.625 x 8.15 (EH 6.5)
CAPA. m3	(252.1 (EH 201.1))
HRT min	(51.5)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4109 A
SERVICE	HF 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE W x L x H | m	5.7 x 5.625 x 8.15 (EH 6.5)
CAPA. m3	(261.3 (EH 208.4))
HRT min	53.4
MAT'L	P.C + FRP

ITEM NO.	W14-T-4110 A
SERVICE	HF 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE W x L x H | m	11.7 x 10.85 x 8.15 (EH 6.5)
CAPA. m3	(1034.6 (EH 825.1))
HRT hr	3.5
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4110 A
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA. W x L x H | m	BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304

MAKER PACKAGE

ITEM NO.	W14-AG-4107 A
SERVICE	HF 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED rpm	(30)
MAT'L	STS316L/STS316L
POWER kW	(15)

ITEM NO.	W14-AG-4108 A
SERVICE	HF 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED rpm	(25)
MAT'L	STS316L/STS316L
POWER kW	(11)

ITEM NO.	W14-AG-4109 A
SERVICE	HF 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED rpm	(25)
MAT'L	STS316L/STS316L
POWER kW	(7.5)

MAKER PACKAGE

ITEM NO.	W14-RD-4110 A
SERVICE	HF 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA. m3/d	(0.067)
MAT'L	STS304/STS304
POWER kW	(2.2)

ITEM NO.	W14-P-4110 A1/A2
SERVICE	HF 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VWF), VORTEX
CAPA. m3/hr.ea x mH	50 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER kW	(30)

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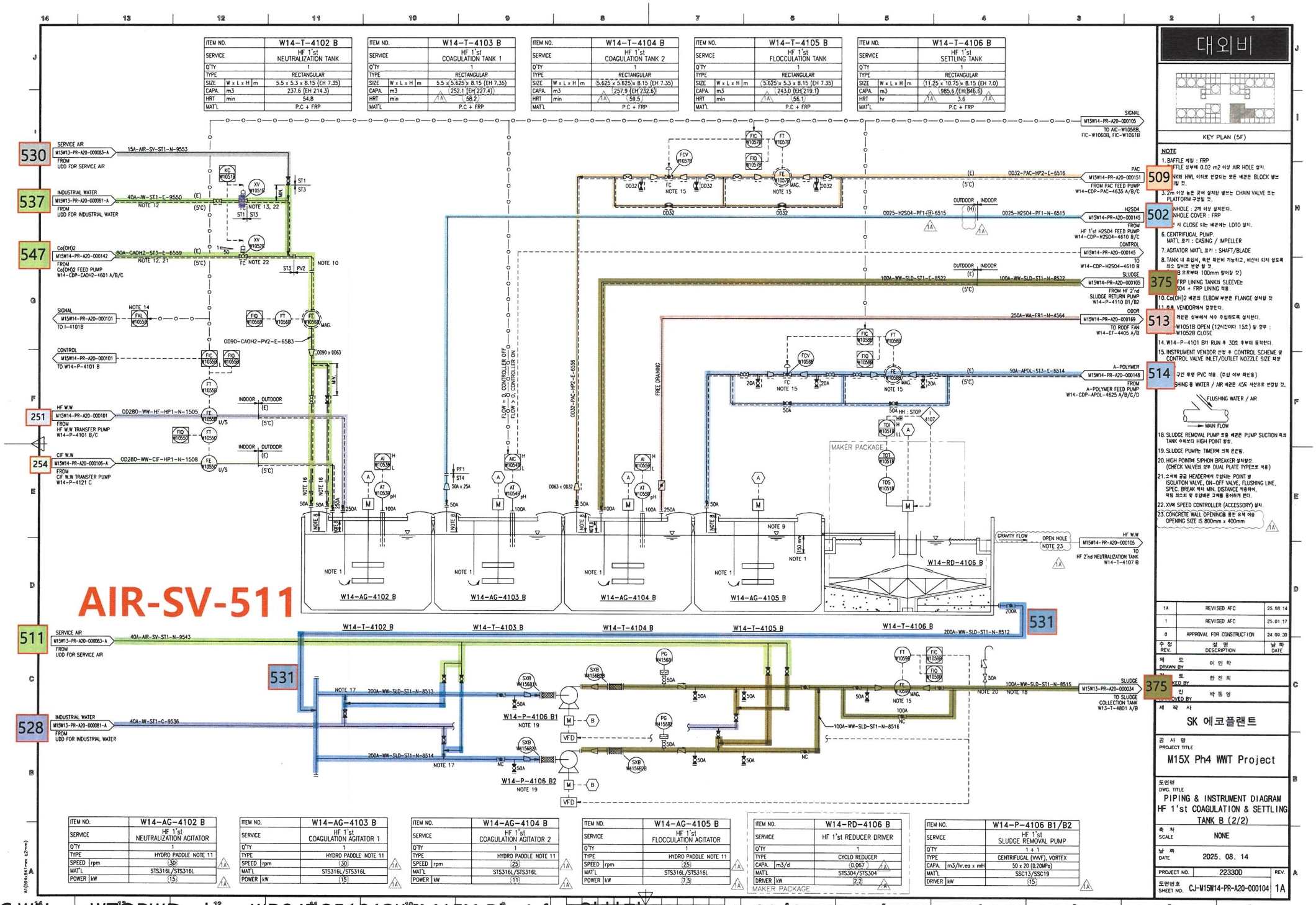
대외비
KEY PLAN (5F)
AIR-SV-511
W14-T-4102 B
W14-T-4103 B
W14-T-4104 B
W14-T-4105 B
W14-T-4106 B
W14-AG-4102 B
W14-AG-4103 B
W14-AG-4104 B
W14-AG-4105 B
W14-RD-4106 B
W14-P-4106 B1
W14-P-4106 B2
MAKER PACKAGE
NOTE
1. BAFFLE 재질 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
2. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치 것.
3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성될 것.
4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
5. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
7. AGITATOR MAT'L 표기 : SHAFT/BLADE
8. TANK 내 유입시, 육안 확인이 가능하고, 바닥이 더지 않도록 의소 깊이로 보완 될 것.
9. 콘크리트로부터 100mm 띄워질 것)
FRP LINING TANK와 SLEEVE는 304 + FRP LINING 적용
10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치될 것
11. 모든 VENDOR에서 검토한다.
12. 라인은 상부에서 시수 주입하도록 설치한다.
W1051B OPEN (12시간마다 1초) 및 경우 :
W1052B CLOSE
14. W14-P-4101 B가 RUN 후 30초 후부터 동작한다.
15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
구간 투명 PVC 적용. (주심 여부 확인필)
FLUSHING 용 WATER / AIR 배관은 45도 사선으로 연결할 것.
18. SLUDGE REMOVAL PUMP 흡입 배관은 PUMP SUCTION 측의 TANK 수위보다 HIGH POINT 할것.
19. SLUDGE PUMP는 TIMER에 의해 운전됨.
20. HIGH POINT에 SIPHON BREAKER 설치할것.
(CHECK VALVE의 경우 DUAL PLATE TYPE으로 적용)
21. 소석회 공급 HEADER에서 주입되는 POINT 별 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 위치 MIN. DISTANCE 적용하여, 적합 조건 및 주입배관 고려를 종합하여 한다.
22. VVH SPEED CONTROLLER (ACCESSORY) 설치.
23. CONCRETE WALL OPENING에 대한 상세 아래 OPENING SIZE IS 800mm x 400mm
KEY PLAN (5F)
1A REVISED AFC 25.08.14
1 REVISED AFC 25.01.17
0 APPROVAL FOR CONSTRUCTION 24.09.30
수정 REV. 설명 DESCRIPTION 날짜 DATE
제도 DRAWN BY 이 인 학
검토 CHECKED BY 한 진 희
승인 APPROVED BY 박 동 영
제작사 SK 에코플랜트
공사명 PROJECT TITLE
M15X Ph4 WWT Project
도면명 DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
HF 1'st COAGULATION & SETTLING TANK B (2/2)
축척 SCALE NONE
날짜 DATE 2025. 08. 14
PROJECT NO. 22330D REV.
도면번호 SHEET NO. CJ-M15W14-PR-A20-000104 1A
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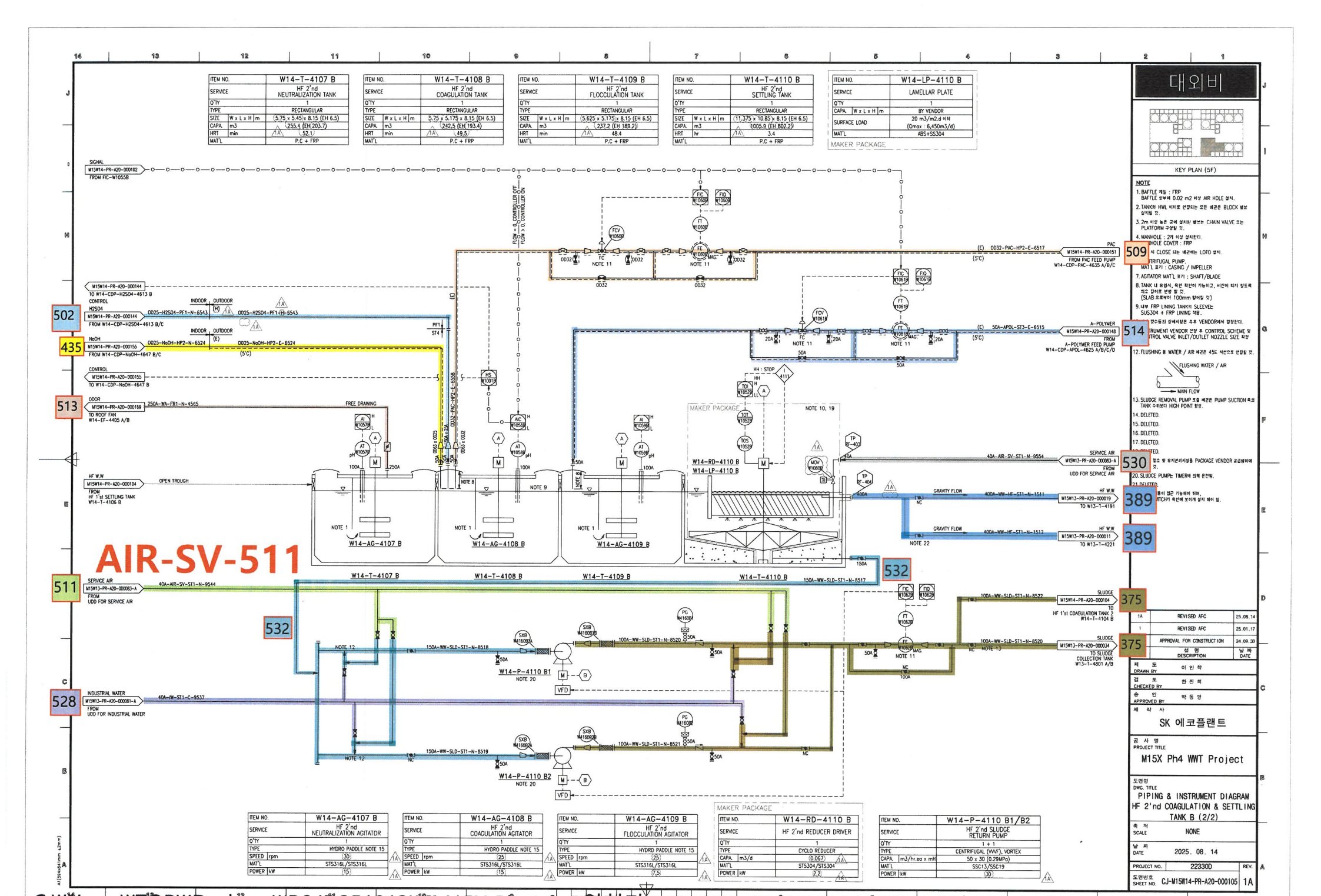
AIR-SV-511
W14-T-4107 B
W14-T-4108 B
W14-T-4109 B
W14-T-4110 B
W14-AG-4107 B
W14-AG-4108 B
W14-AG-4109 B
W14-RD-4110 B
W14-LP-4110 B
W14-P-4110 B1
W14-P-4110 B2
MAKER PACKAGE
대외비
SK 에코플랜트
M15X Ph4 WWT Project
PIPING & INSTRUMENT DIAGRAM
HF 2'nd COAGULATION & SETTLING TANK B (2/2)
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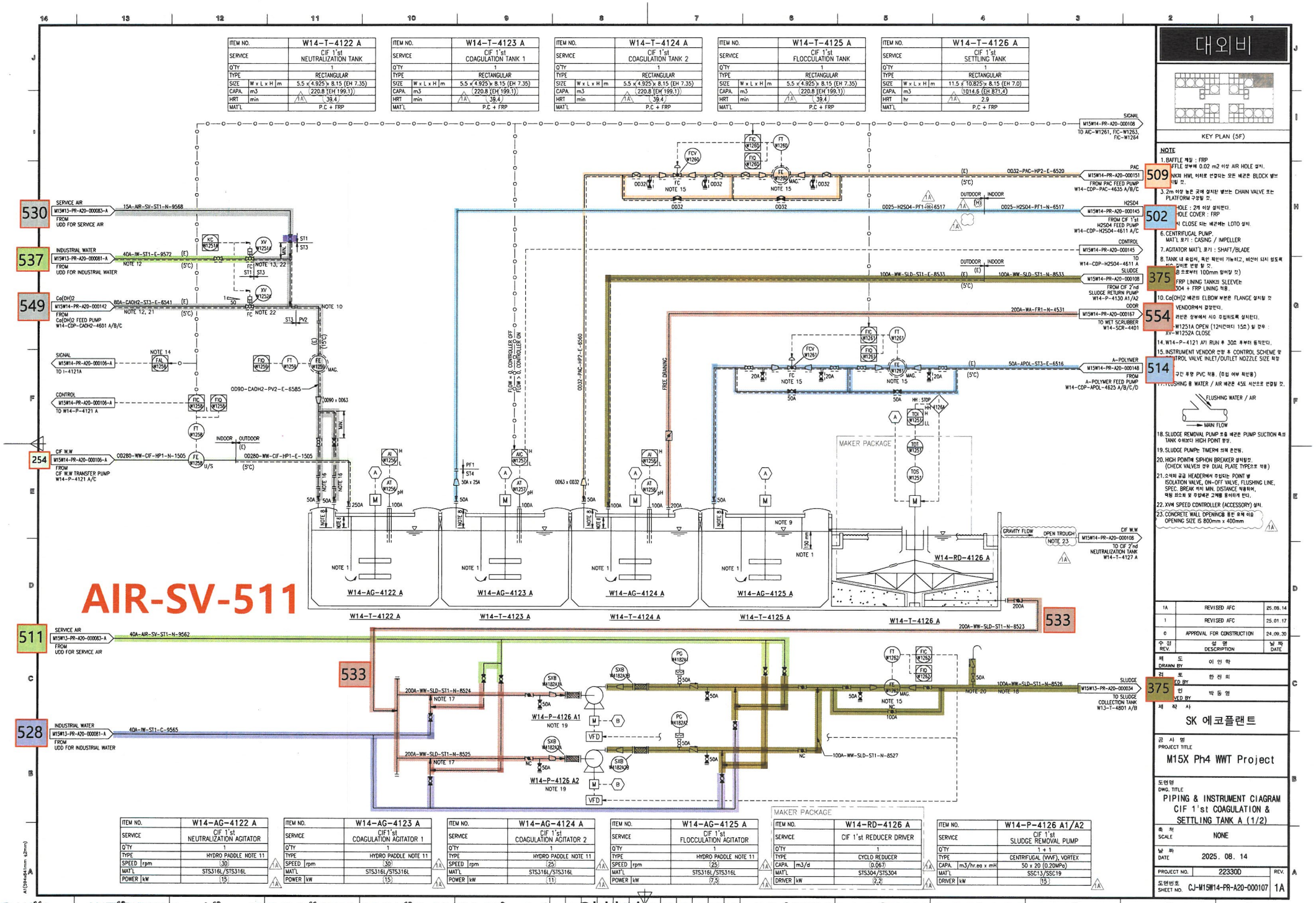

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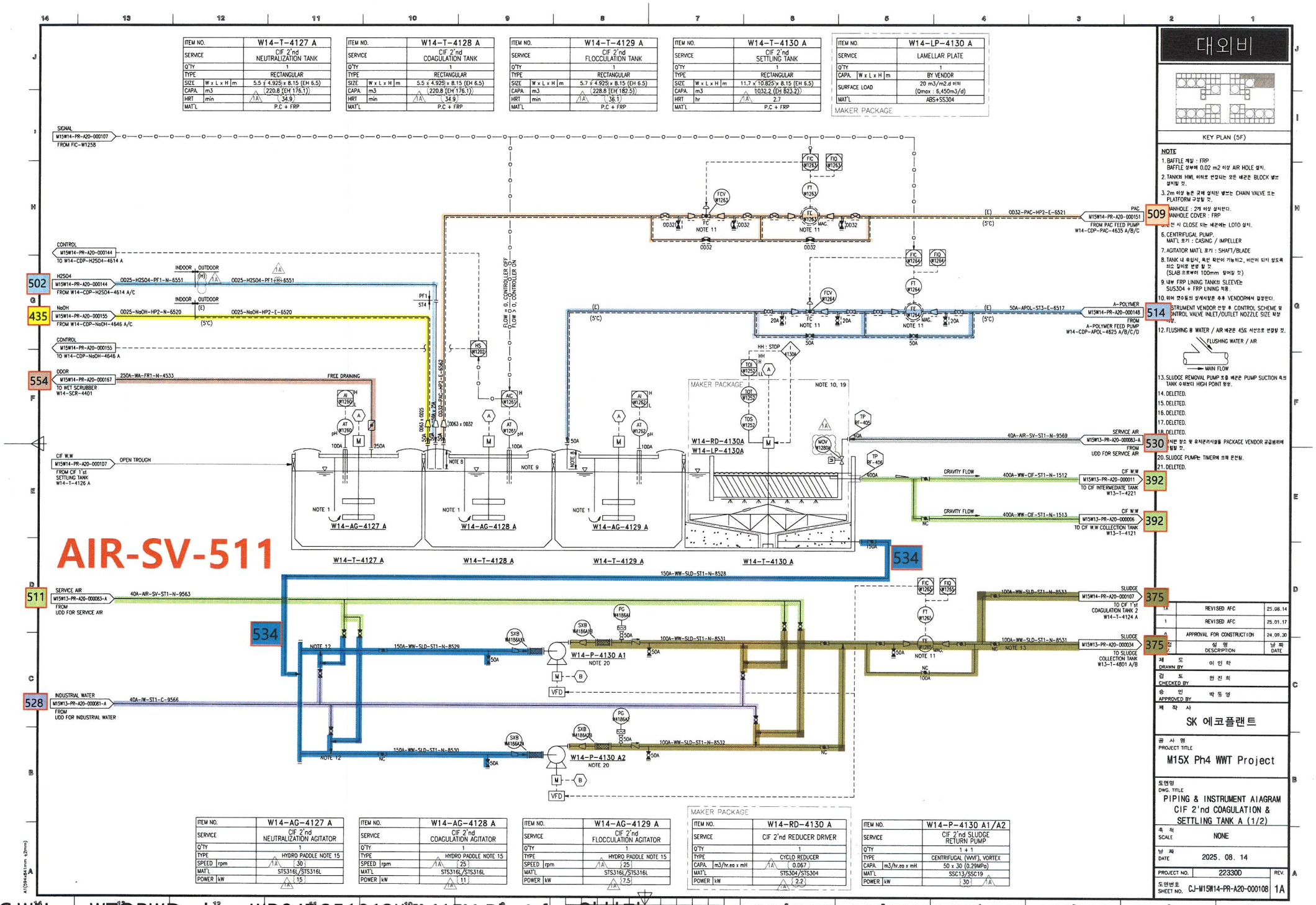

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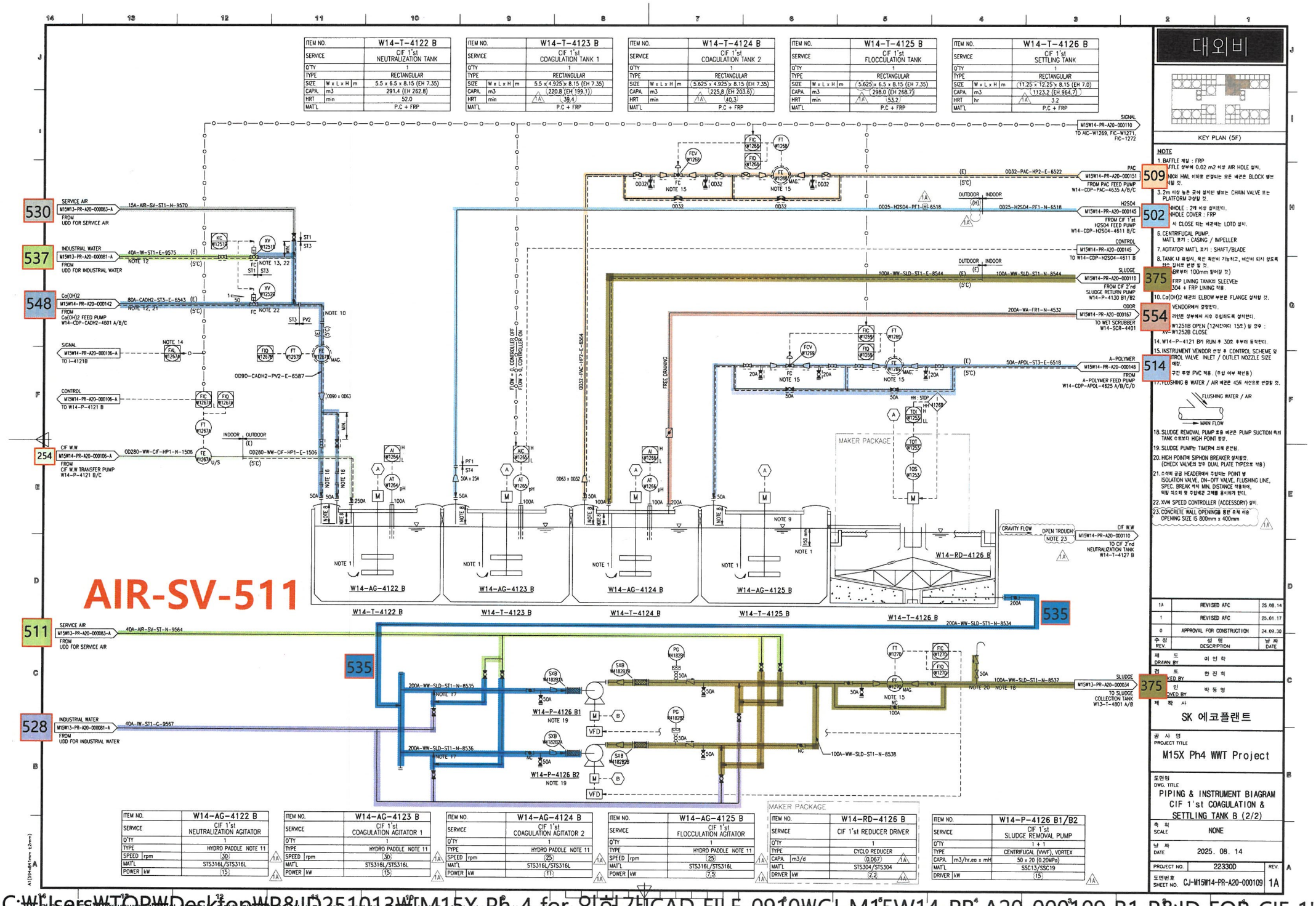
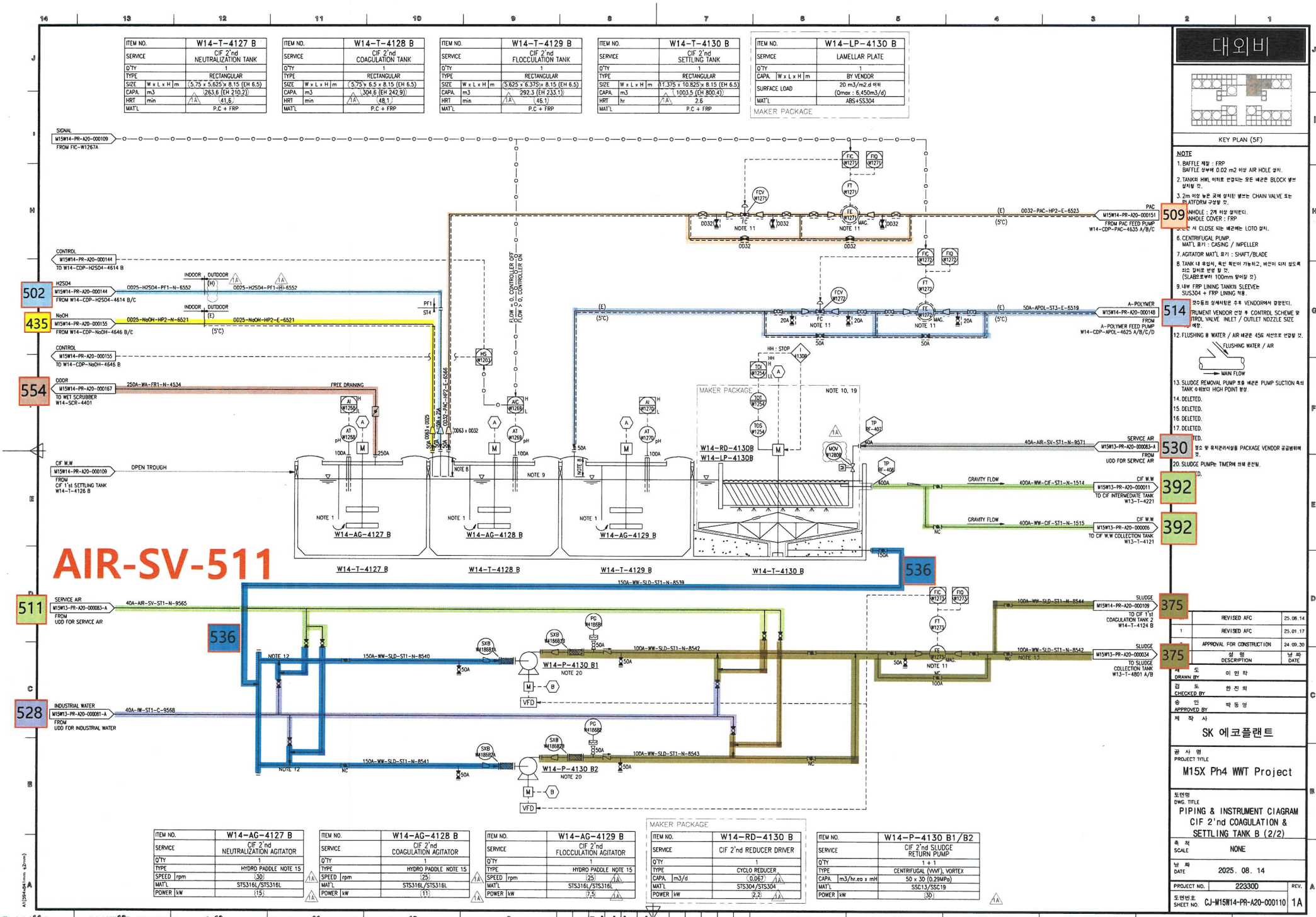

대외비
KEY PLAN (5F)
AIR-SV-511
W14-T-4122 B
CIF 1'st NEUTRALIZATION TANK
W14-T-4123 B
CIF 1'st COAGULATION TANK 1
W14-T-4124 B
CIF 1'st COAGULATION TANK 2
W14-T-4125 B
CIF 1'st FLOCCULATION TANK
W14-T-4126 B
CIF 1'st SETTLING TANK
W14-AG-4122 B
W14-AG-4123 B
W14-AG-4124 B
W14-AG-4125 B
W14-RD-4126 B
W14-P-4126 B1
W14-P-4126 B2
MAKER PACKAGE
NOTE
1. BAFFLE 재질 : FRP
2. BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
5. 사람이 CLOSE 되는 배관에는 LOTO 설치.
6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
7. AGITATOR MAT'L 표기 : SHAFT/BLADE
8. TANK 내 유입식, 확산 확인이 가능하고, 배관이 더치 않도록 하수 깊이로 변경 할 것.
9. 바닥으로부터 100mm 띄워질 것)
10. FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
11. Ca(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것.
12. 제작은 VENDOR에서 검토한다.
13. 라인은 상부에서 시수 주입되도록 설치한다.
FCV-W1251B OPEN (12시간마다 15초) 및 경우 : XV-W1252B CLOSE
14. W14-P-4121 B가 RUN 후 30초 후부터 동작한다.
15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 변경.
16. 배관은 주로 PVC 적용. (주요 이부 확인됨)
17. FLUSHING 용 WATER / AIR 배관은 45도 사선으로 변경할 것.
18. SLUDGE REMOVAL PUMP 흡입 배관은 PUMP SUCTION 측의 TANK 수위보다 HIGH POINT 형성.
19. SLUDGE PUMP는 TIMER에 의해 운전됨.
20. HIGH POINT에 SIPHON BREAKER 설치할것. (CHECK VALVE의 경우 DUAL PLATE TYPE으로 적용)
21. 소석회 공급 HEADER에서 주입되는 POINT 별 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 위치 MIN. DISTANCE 적용하여, 적절 주소의 별 주입배관 구체를 용이하게 한다.
22. XYM SPEED CONTROLLER (ACCESSORY) 설치.
23. CONCRETE WALL OPENING을 통한 유체 이송 OPENING SIZE IS 800mm x 400mm
FLUSHING WATER / AIR
MAIN FLOW
REVISED AFC 25.08.14
REVISED AFC 25.01.17
APPROVAL FOR CONSTRUCTION 24.09.30
SK 에코플랜트
M15X Ph4 WWT Project
PIPING & INSTRUMENT DIAGRAM CIF 1'st COAGULATION & SETTLING TANK B (2/2)
NONE
2025. 08. 14
22330D
CJ-M15W14-PR-A20-000109
1A
C:₩Users₩TCPWDesktop₩P&ID251013₩[M15X-Ph 4 for 외부기관]CAD FILE 0910₩CJ-M15W14-PR-A20-000109 R1 P&ID FOR CIF 1'st



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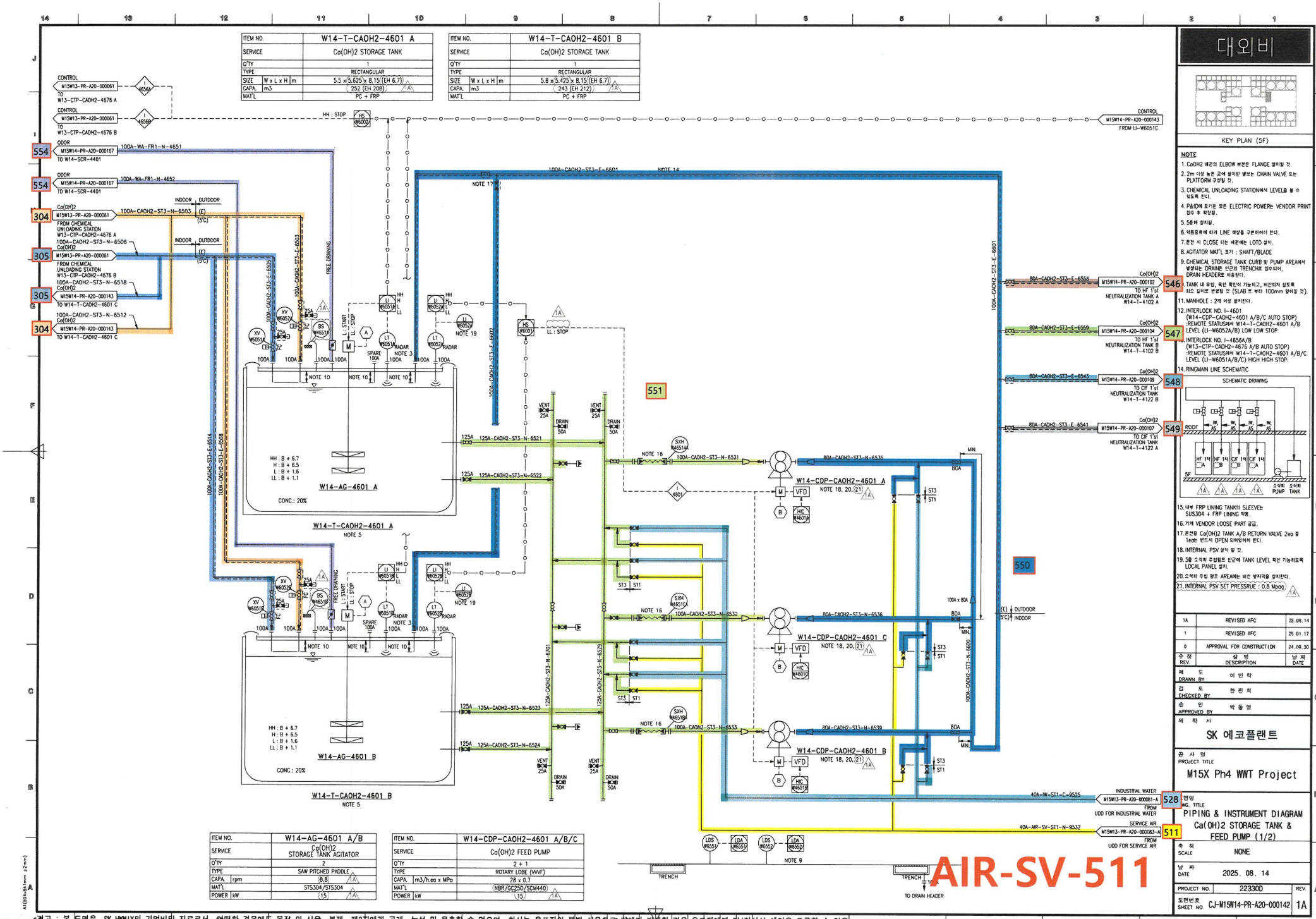
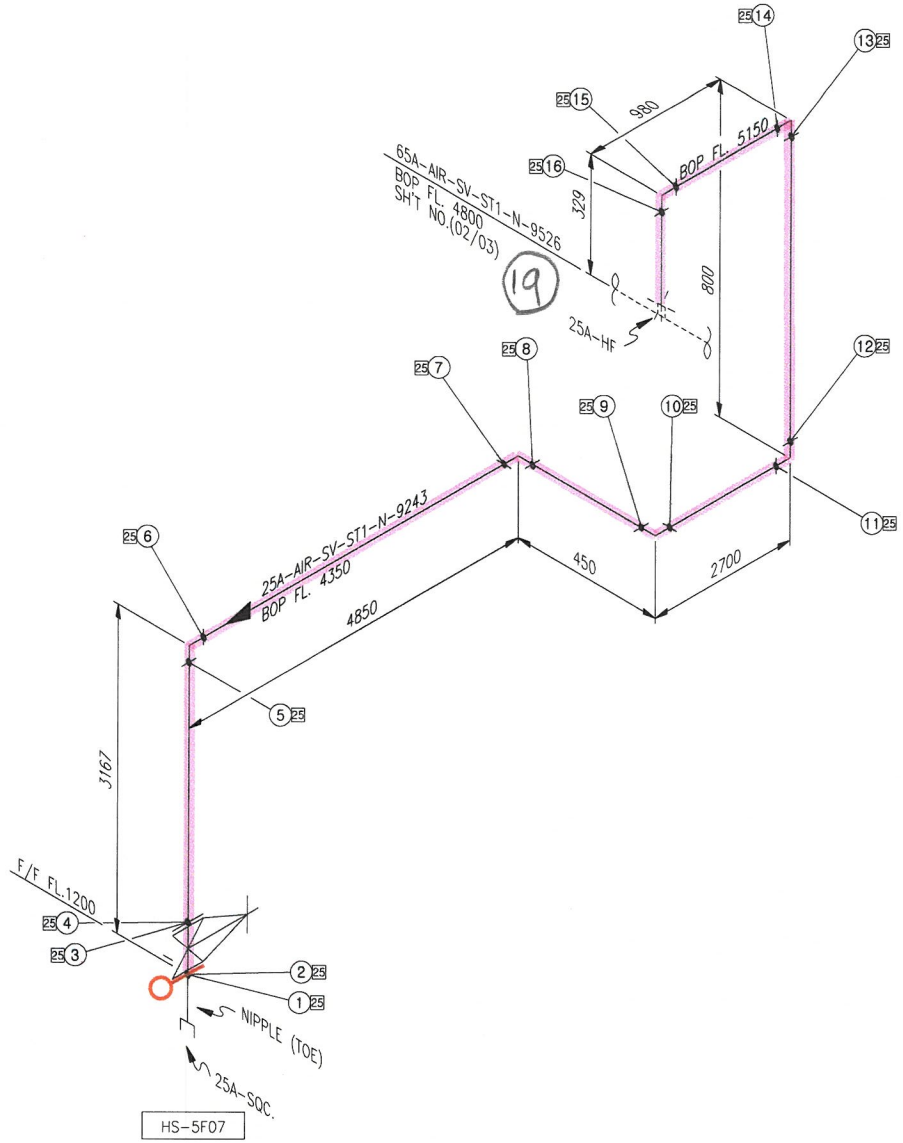


경고 : 본 도면은 SK hynix의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 수성 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 인해서 발생하는 모든 민/형사상 책임을 묵과할 수 없음!!



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	13.5M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	6EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (1)

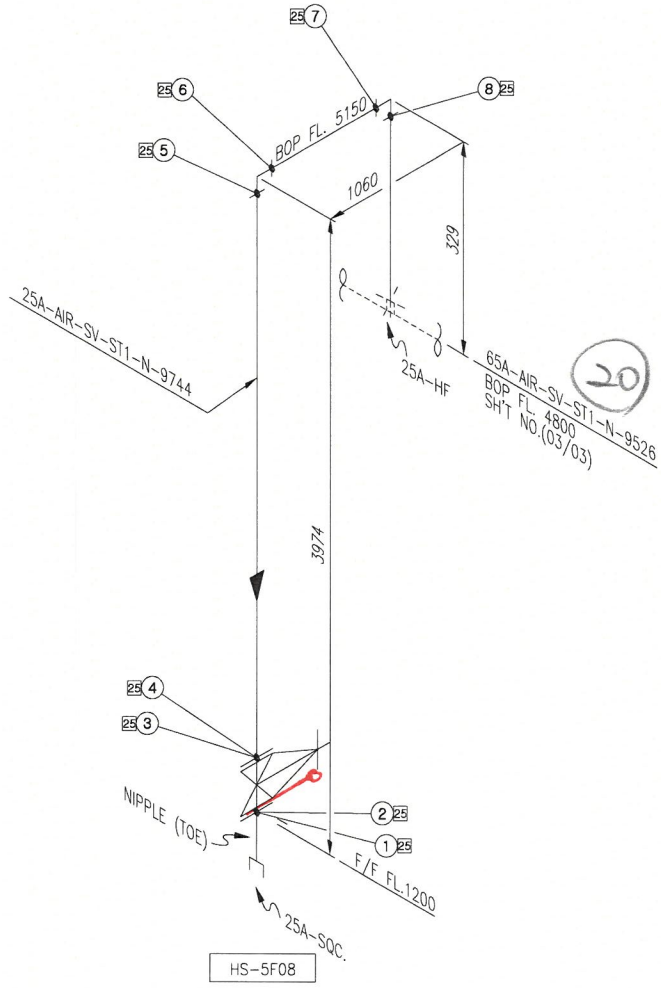
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050004-D
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9743
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	5.8M	PIPE, STS 304TP, ERW, PE – SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE – SCH 40S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (2)

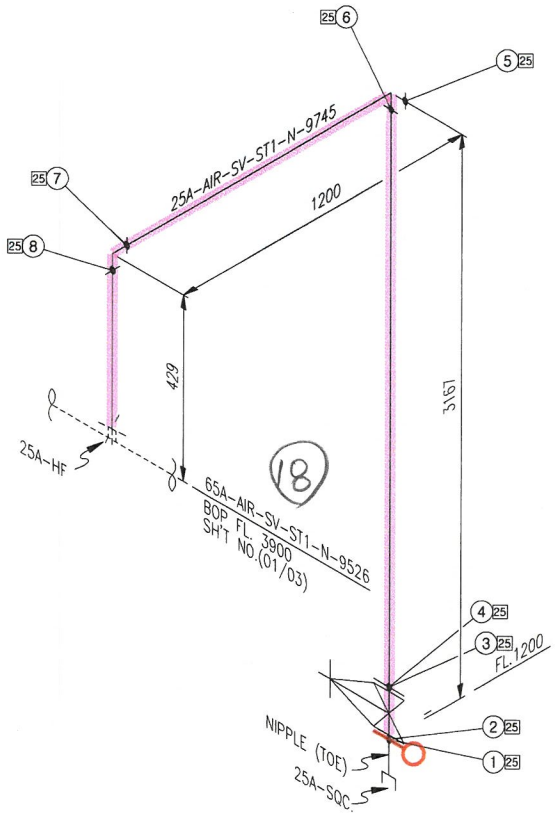
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	10	60
			OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050007-D
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 25A-AIR-SV-ST1-N-9744

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0



HS-5F06

<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	5.1M	PIPE, STS 304TP, ERW, PE - SCH 10S
NP1	25A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304, BB / OS&Y, HANDWHEEL
F	25A	2EA	FLG SO, KS 10K, RF, STS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	25A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
SQC	25A	1EA	QUICK COUPLING(MALE)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (3)

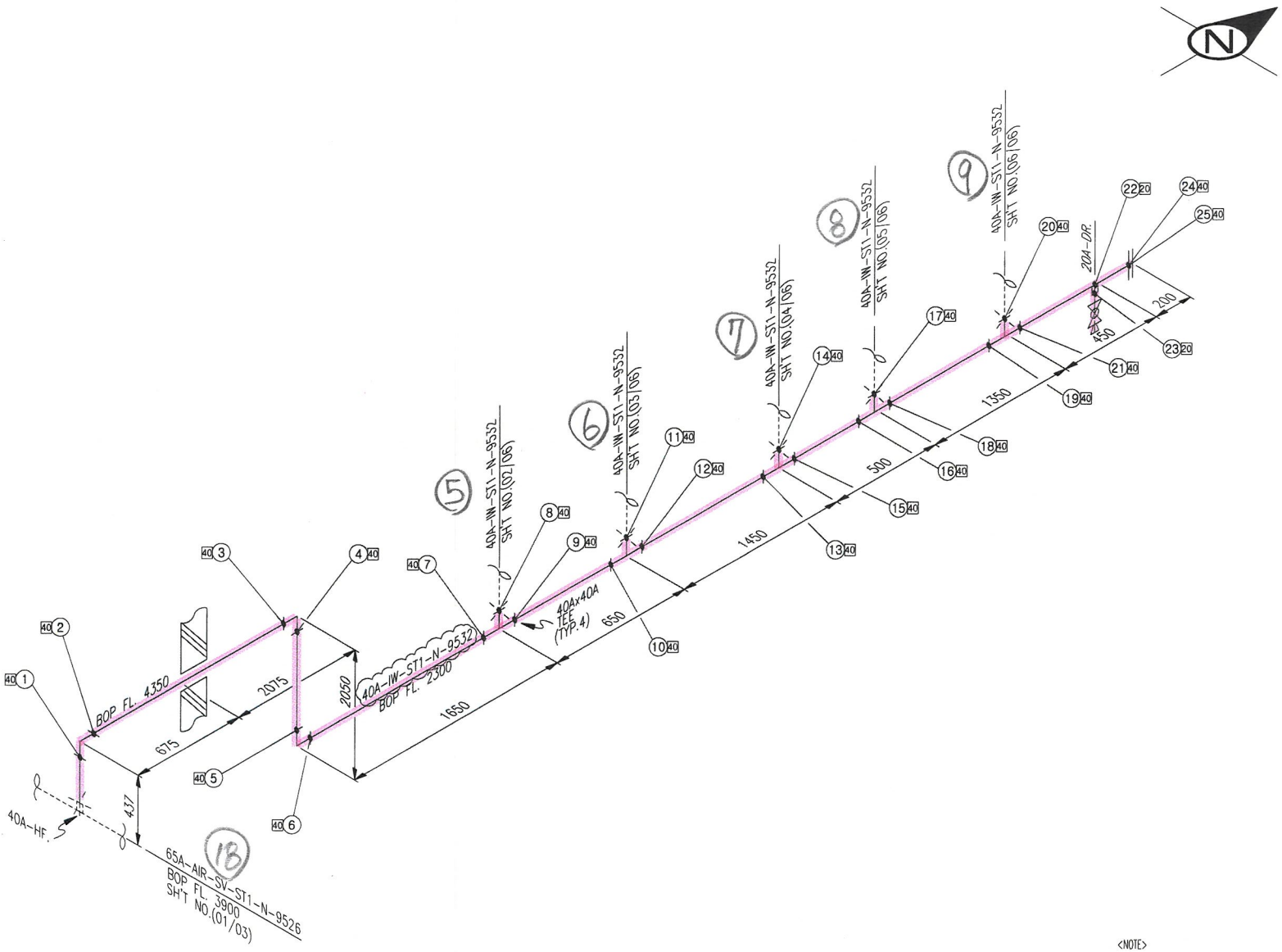
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY AIR-SV
JOB NO. 22330D
LINE NO. 25A-AIR-SV-ST1-N-9745
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	9.5M	PIPE, STS 304TP, ERW, PE – SCH 10S
NP1	20A	1EA	NIPPLE, STS 304TP, ERW, T0E – SCH 40S
E90	40A	3EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
T	40A	5EA	TEE EQUAL, STSF304, SW – SCH 10S
HF	20A	1EA	COUPLING HALF, STSF304, SW – SCH 10S
PLG	20A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
F	40A	1EA	FLG SO, KS 10K, RF, STS304
FB	40A	1EA	FLG BLIND, KS 10K, RF, STS304
G	40A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (4)

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

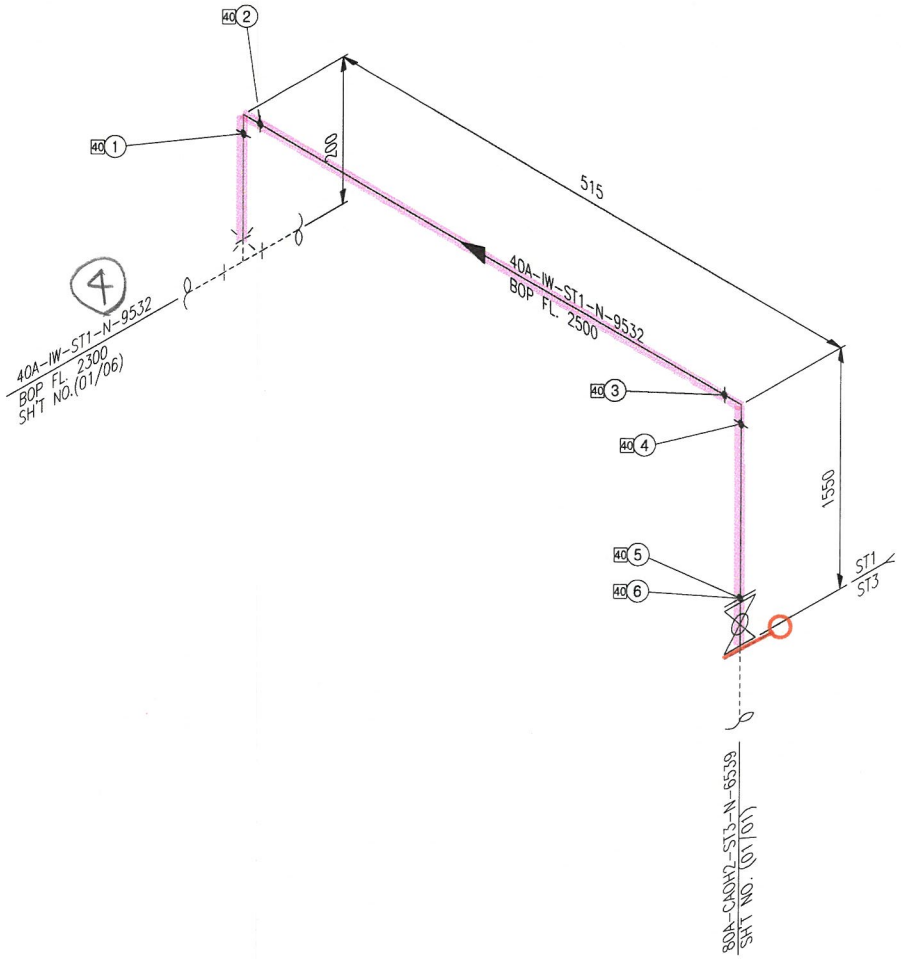
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C
REF. P & ID CJ-M15W14-PR-A20-000142

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
D	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9532

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 06
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	2.3M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	40A	2EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
VB	40A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY–STSF304 /
			TRIM–STS304 / SEAT–PTFE, FB, LEVER
F	40A	1EA	FLG SO, KS 10K, RF, STS304
G	40A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix 5

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	10	60
			OPERATING	7.5	AMB

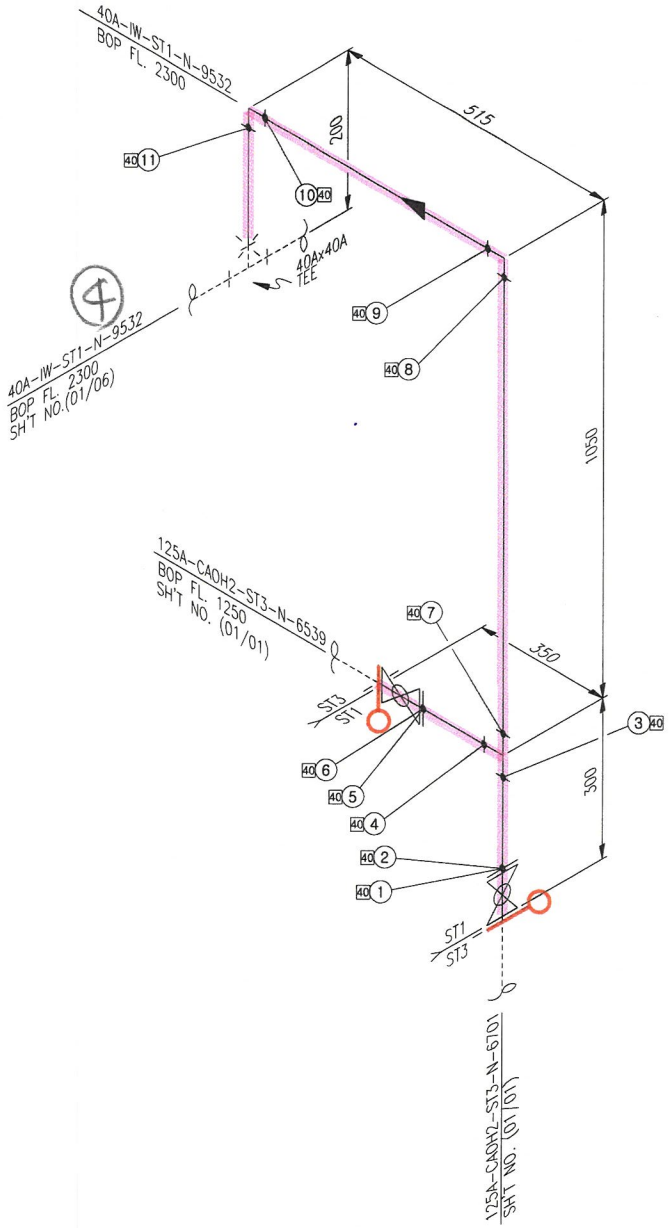
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C
REF. P & ID CJ-M15W14-PR-A20-000142

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9532

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 02 / 06
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	2.4M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	2EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	1EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	2EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	2EA	FLG SO, KS 10K, RF, STS304
G	40A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

CLIENT SK hynix (6)

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

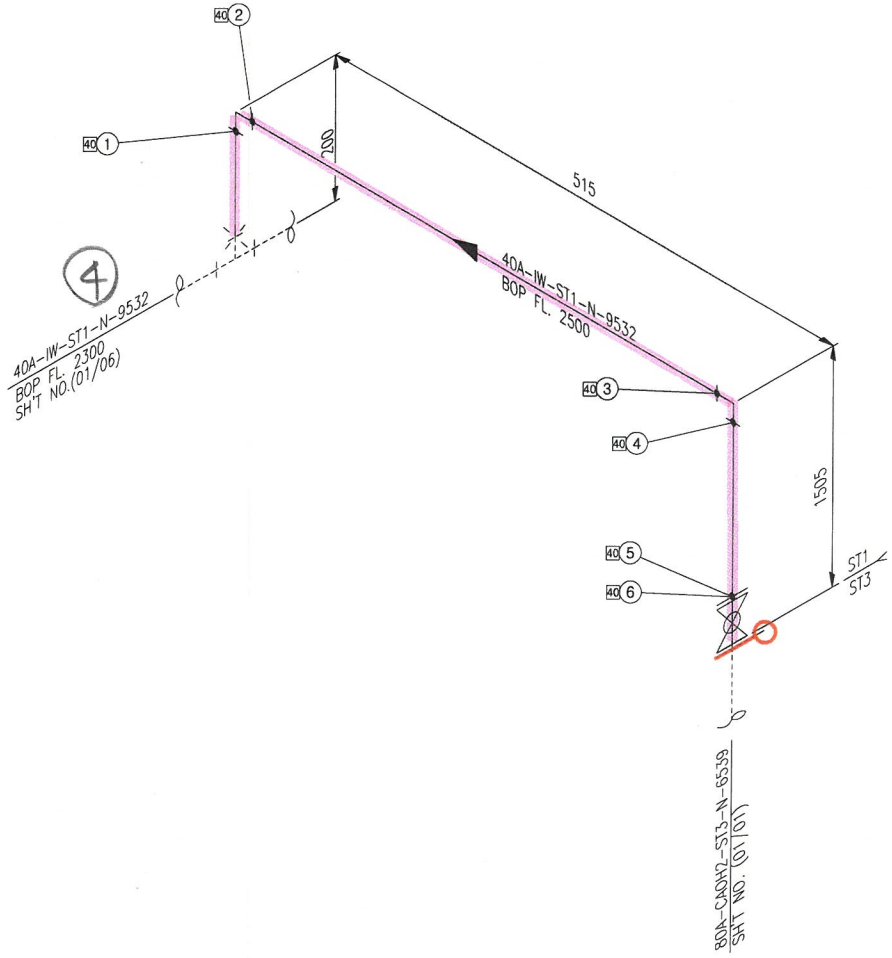
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C
REF. P & ID CJ-M15W14-PR-A20-000142

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9532

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 03 / 06
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	2.3M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	40A	2EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
VB	40A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	1EA	FLG SO, KS 10K, RF, STS304
G	40A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	10	60
			OPERATING	7.5	AMB

AREA	W14
UNIT	5FA
REF. PLAN DWG.	CJ-M15W14-PI-A30-050005-C
REF. P & ID	CJ-M15W14-PR-A20-000142

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant

PROJECT NAME M15X Ph4 Project

COMMODITY AIR-SV

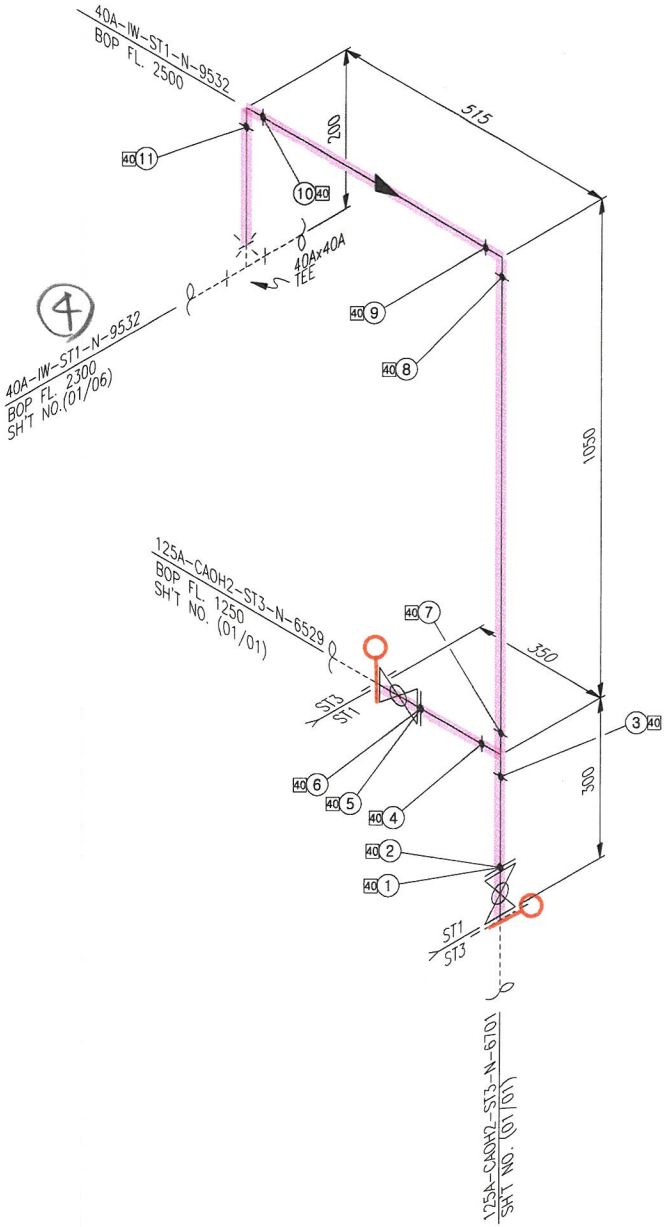
JOB NO. 22330D

LINE NO. 40A-AIR-SV-ST1-N-9532

SH'T NO. 04 / 06

REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	2.6M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	2EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	1EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	3EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	3EA	FLG SO, KS 10K, RF, STS304
G	40A	3EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	3SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C
REF. P & ID CJ-M15W14-PR-A20-000142

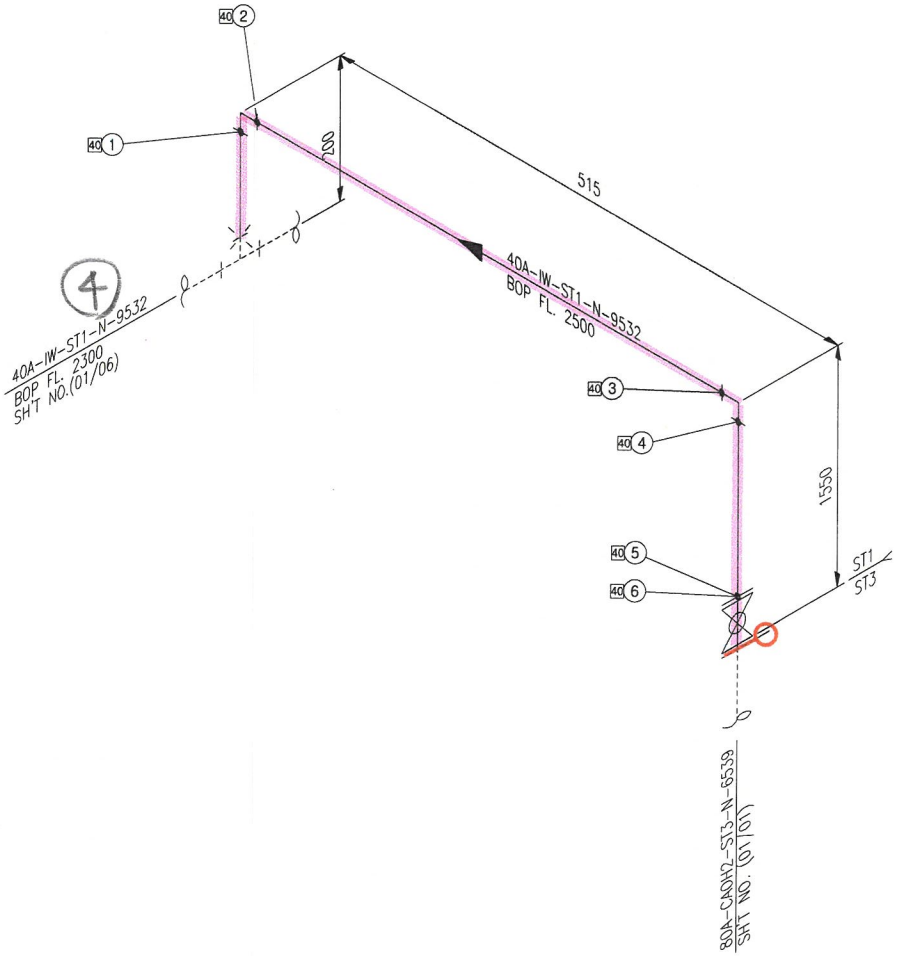
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9532

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 05 / 06
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	2.3M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	40A	2EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
VB	40A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY–STSF304 /
			TRIM–STS304 / SEAT–PTFE, FB, LEVER
F	40A	1EA	FLG SO, KS 10K, RF, STS304
G	40A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (9)

INSULATION	CODE	N
	THICK.	–

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

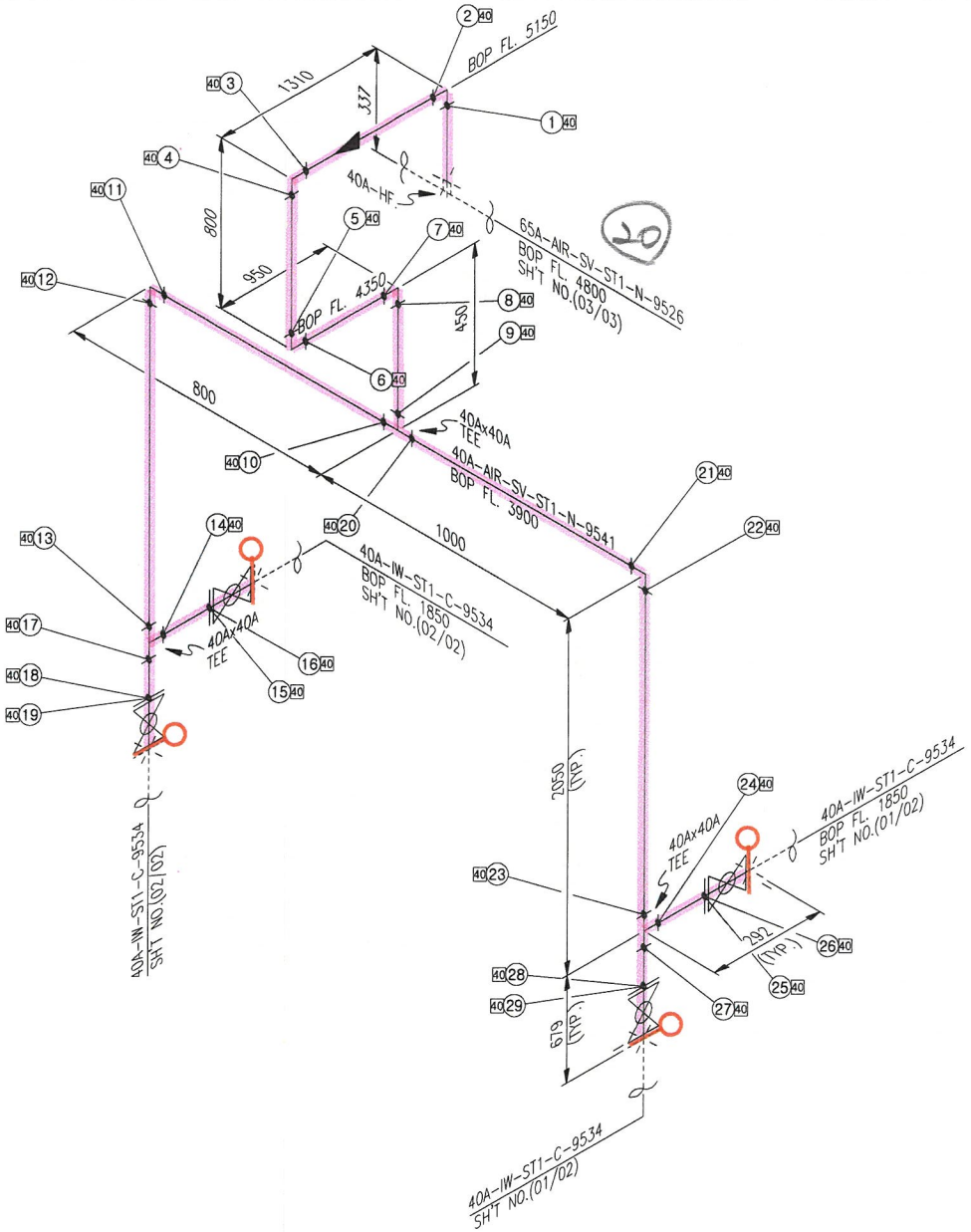
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C
REF. P & ID CJ-M15W14-PR-A20-000142

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9532

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 06 / 06
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	11.7M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	6EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	4EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	4EA	FLG SO, KS 10K, RF, STS304
G	40A	8EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	8SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050007-D
REF. P & ID CJ-M15W14-PR-A20-000102

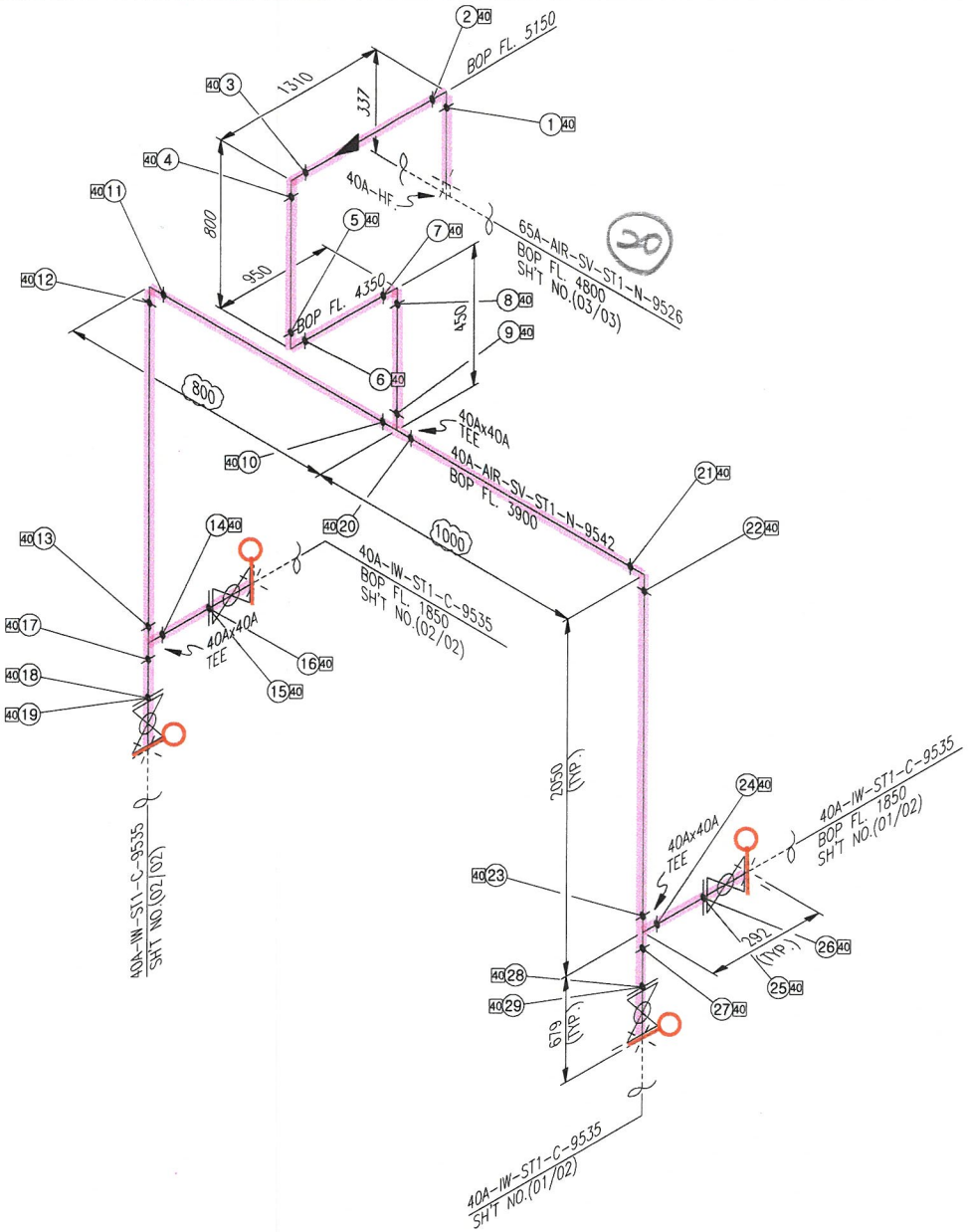
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9541

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

CLIENT SK hynix

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	11.7M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	6EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	4EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	4EA	FLG SO, KS 10K, RF, STS304
G	40A	8EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	8SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

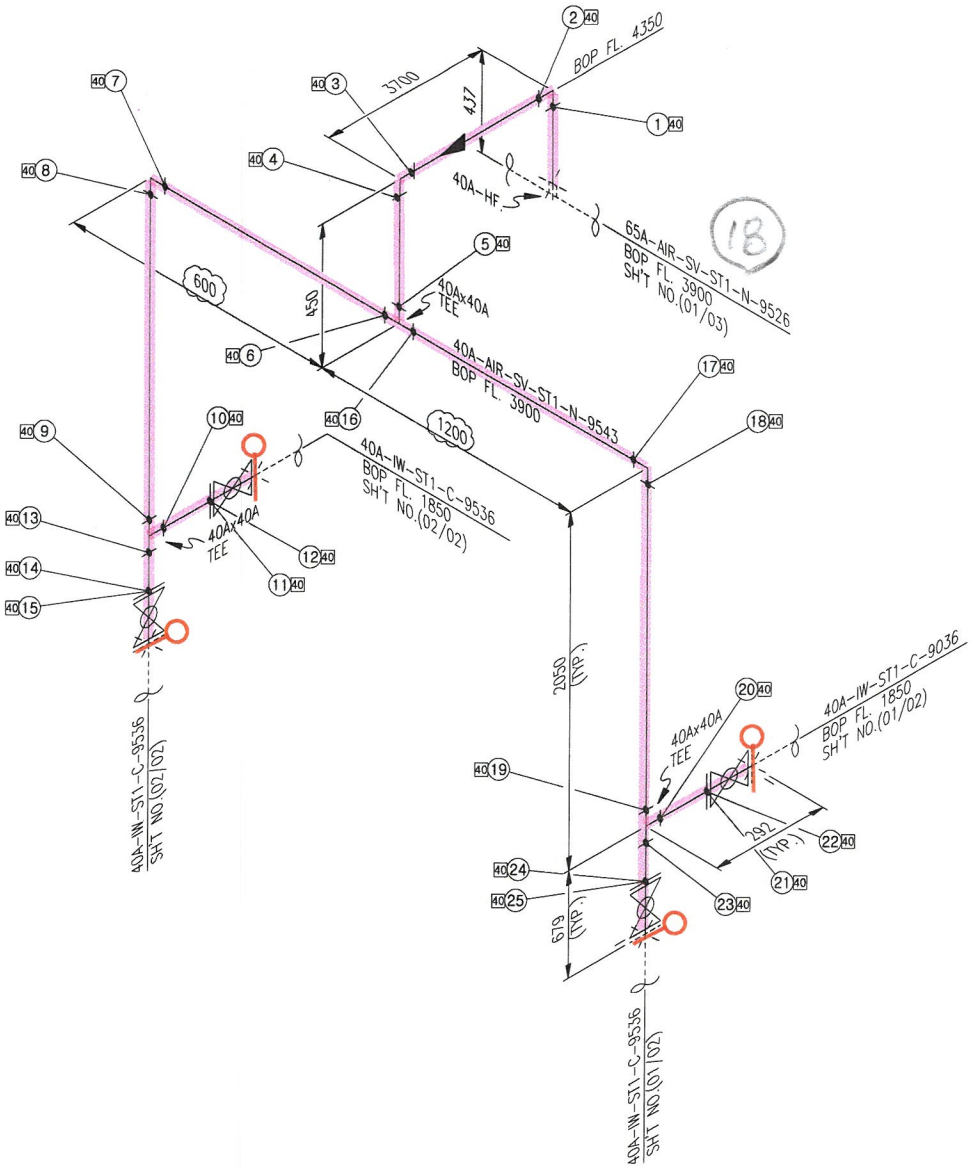
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050007-D
REF. P & ID CJ-M15W14-PR-A20-000103

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9542

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	12.5M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	4EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	4EA	FLG SO, KS 10K, RF, STS304
G	40A	8EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	8SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050006-C
REF. P & ID CJ-M15W14-PR-A20-000104

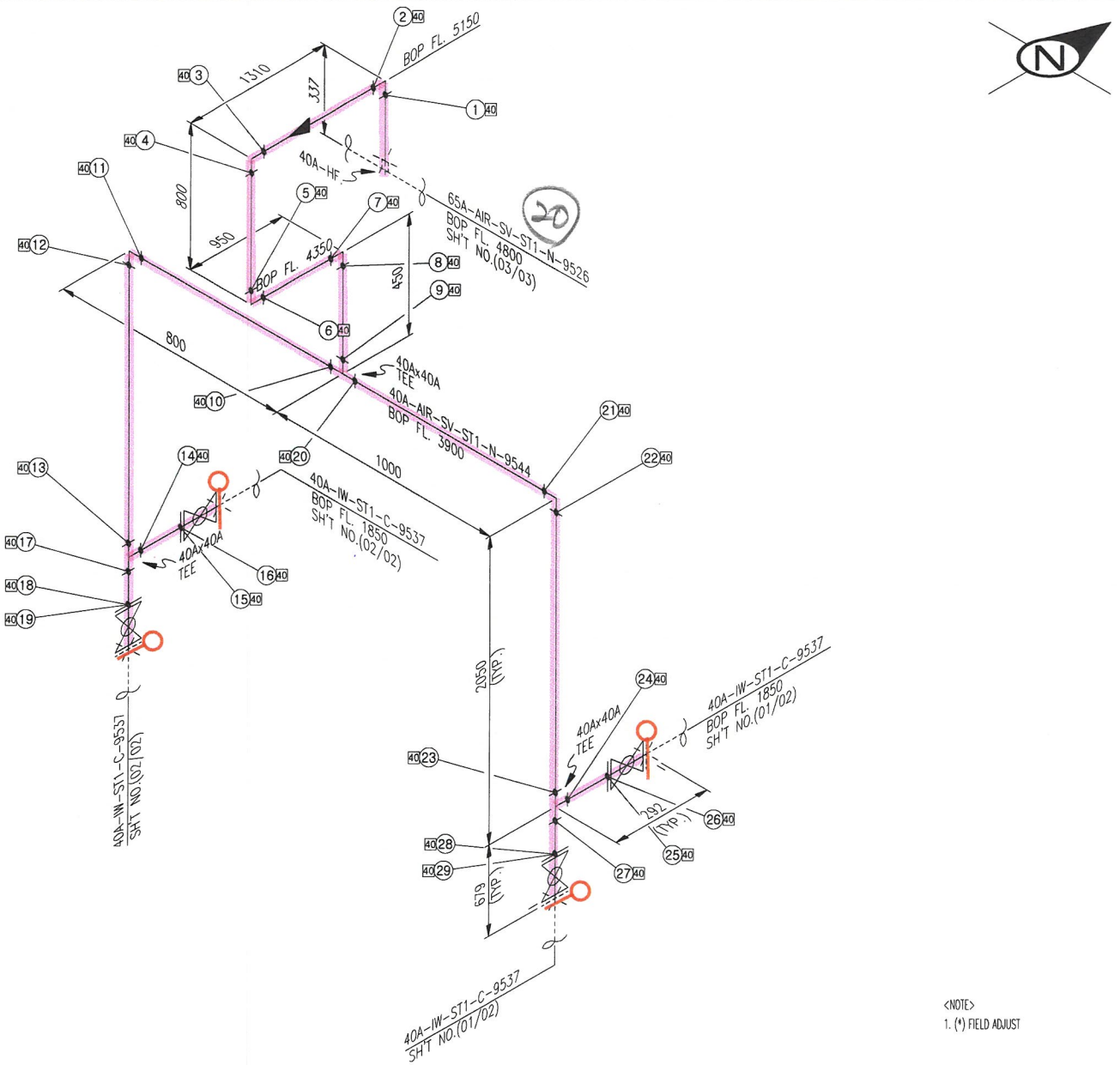
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CLIENT SK hynix

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9543

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	11.7M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	6EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	4EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	4EA	FLG SO, KS 10K, RF, STS304
G	40A	8EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	8SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

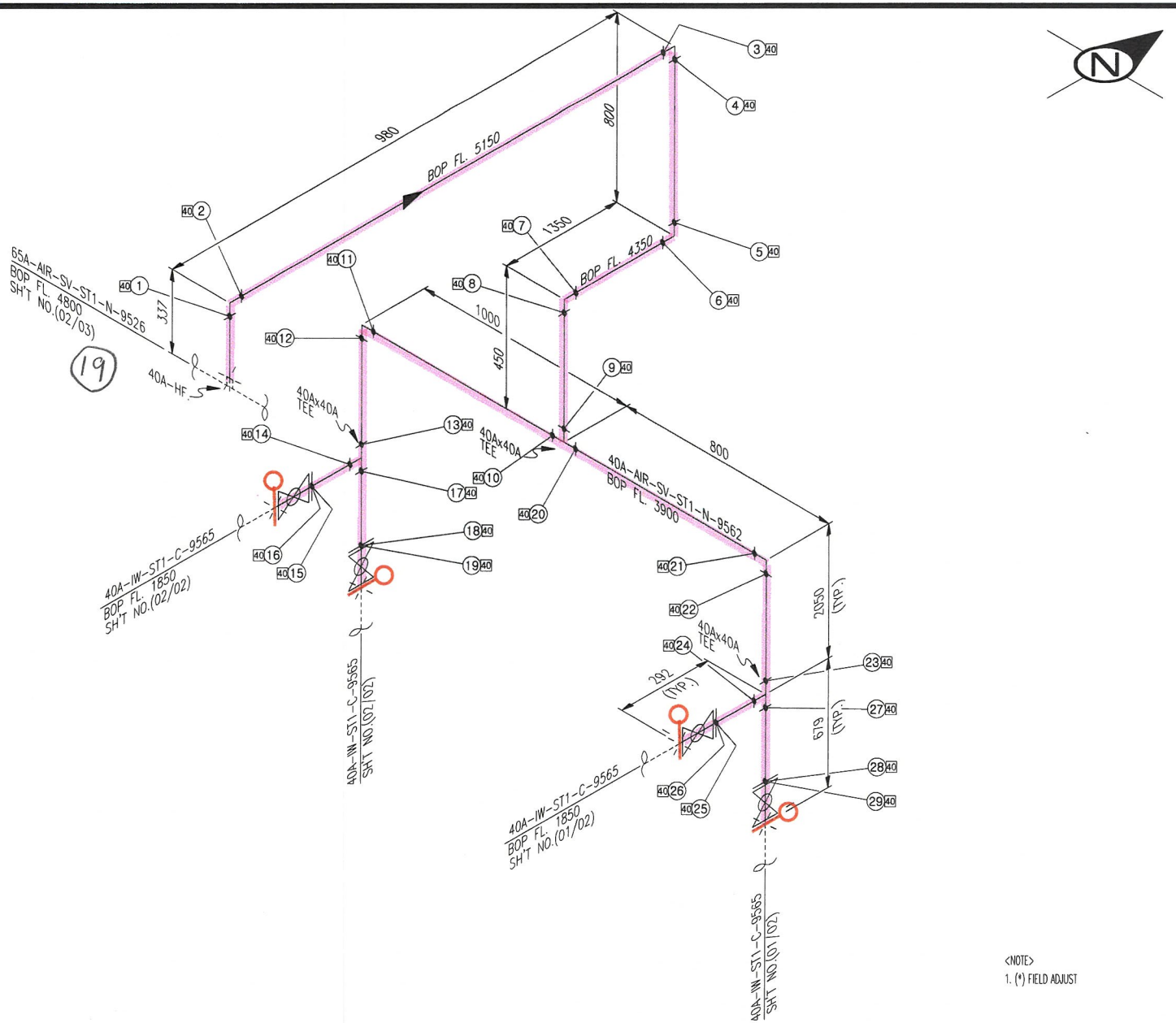
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050008-D
REF. P & ID CJ-M15W14-PR-A20-000105

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9544

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	11.8M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	6EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	4EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	4EA	FLG SO, KS 10K, RF, STS304
G	40A	8EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	8SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (14)

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

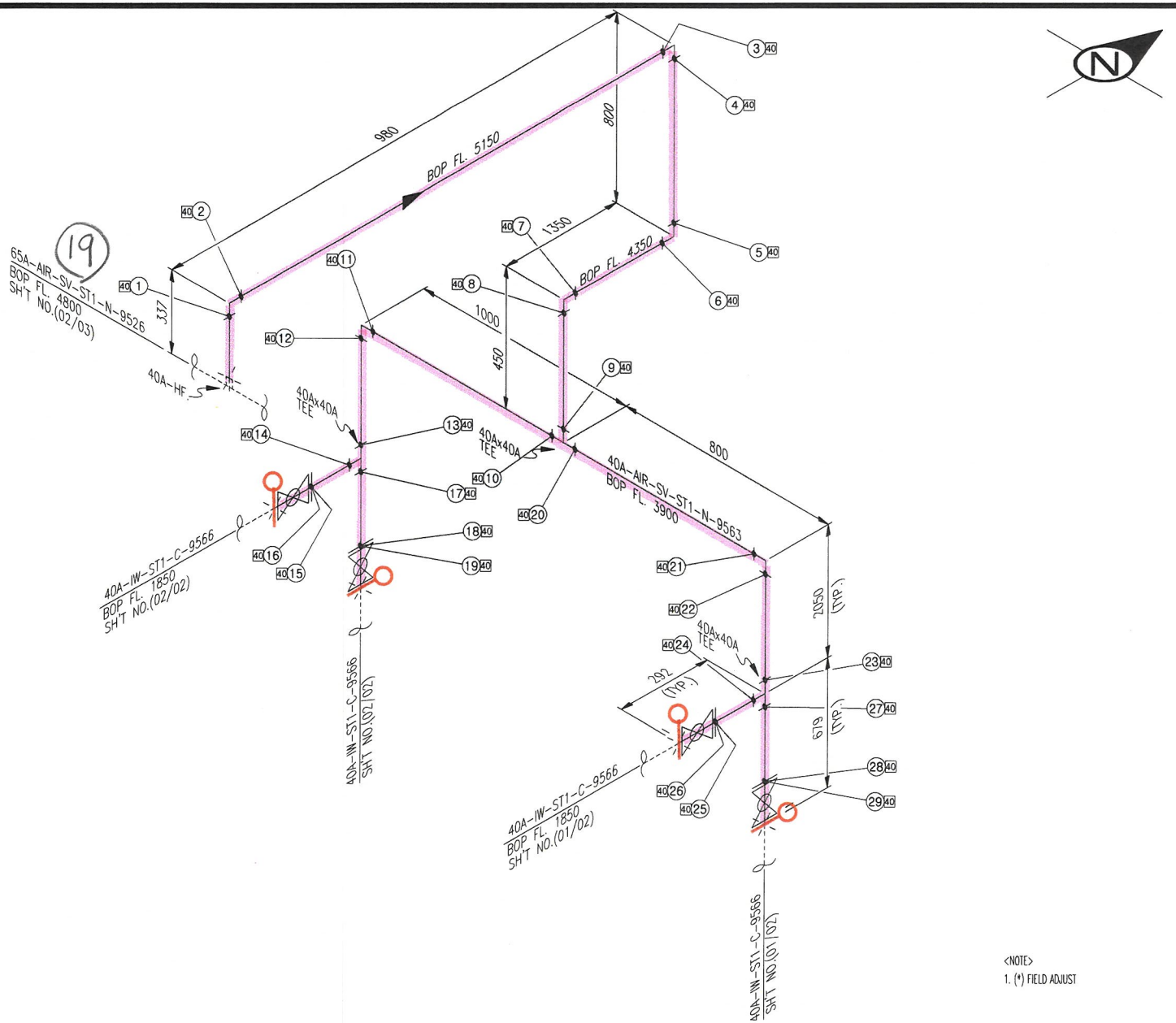
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050004-D
REF. P & ID CJ-M15W14-PR-A20-000107

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동엽

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9562

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



N

<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	11.8M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	40A	6EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW - SCH 10S
VB	40A	4EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	4EA	FLG SO, KS 10K, RF, STS304
G	40A	8EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	8SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90' ELBOW : E90 * 45' ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (15)

INSULATION	CODE	N
	THICK.	-

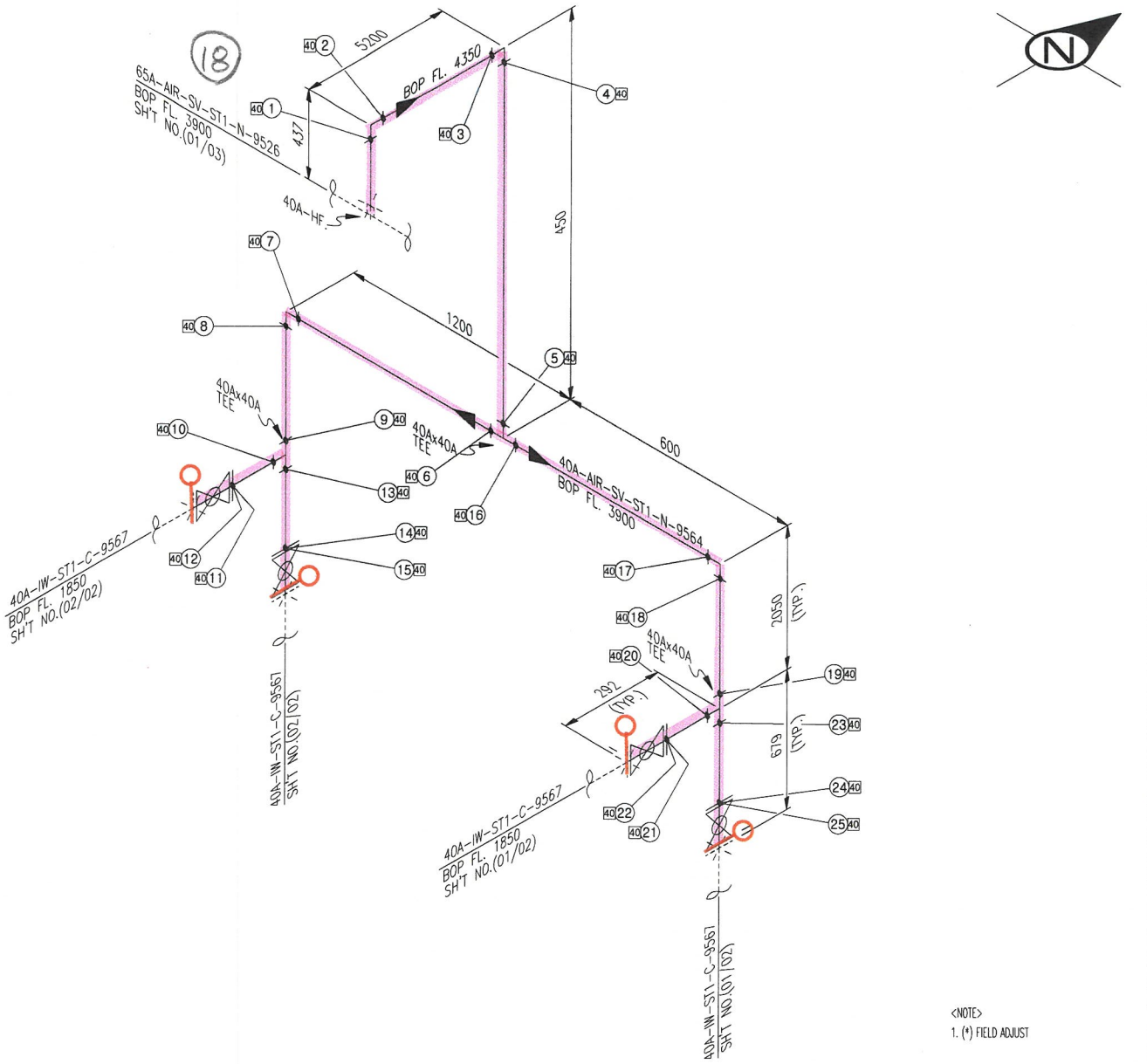
CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050004-D
REF. P & ID CJ-M15W14-PR-A20-000108

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9563
PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	14.0M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW – SCH 10S
VB	40A	4EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	4EA	FLG SO, KS 10K, RF, STS304
G	40A	8EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	8SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (16)

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	10	60
			OPERATING	7.5	AMB

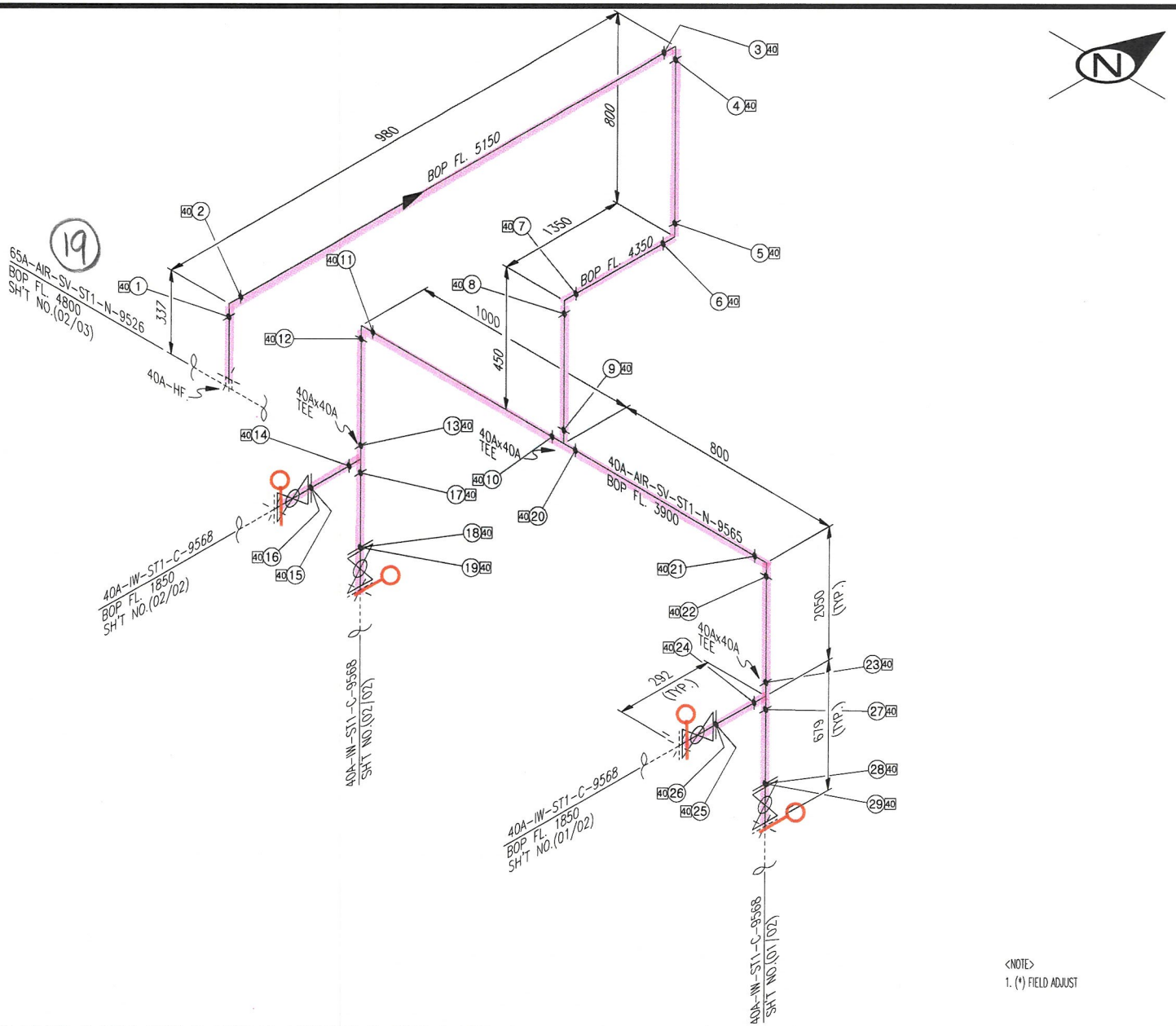
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050005-C
REF. P & ID CJ-M15W14-PR-A20-000109

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 40A-AIR-SV-ST1-N-9564

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	40A	11.8M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	40A	6EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW – SCH 10S
VB	40A	4EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	40A	4EA	FLG SO, KS 10K, RF, STS304
G	40A	8EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	8SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix (17)

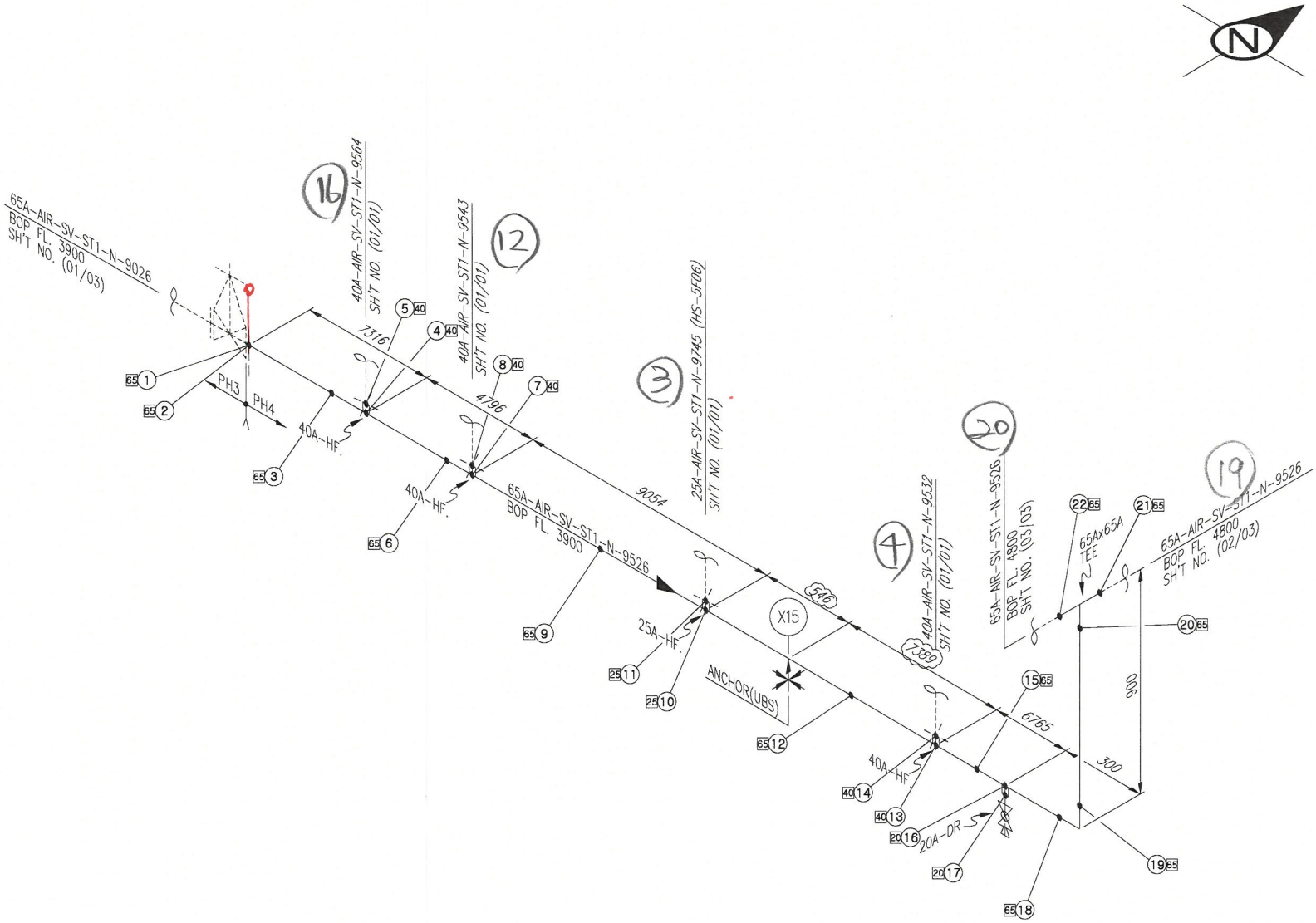
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	10	60
			OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050005-D
REF. P & ID CJ-M15W14-PR-A20-000110

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY AIR-SV
JOB NO. 22330D
LINE NO. 40A-AIR-SV-ST1-N-9565
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. (*) FIELD ADJUST

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	65A	37.0M	PIPE, STS 304TP, ERW, BE - SCH 10S
NP1	20A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	65A	1EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
T	65A	1EA	TEE EQUAL, STS304, ERW, BE - SCH 10S
HF	40A	3EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	25A	1EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	20A	1EA	COUPLING HALF, STSF304, SW - SCH 10S
PLG	20A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	65A	1EA	FLG SO, KS 10K, RF, STS304
G	65A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	65A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix

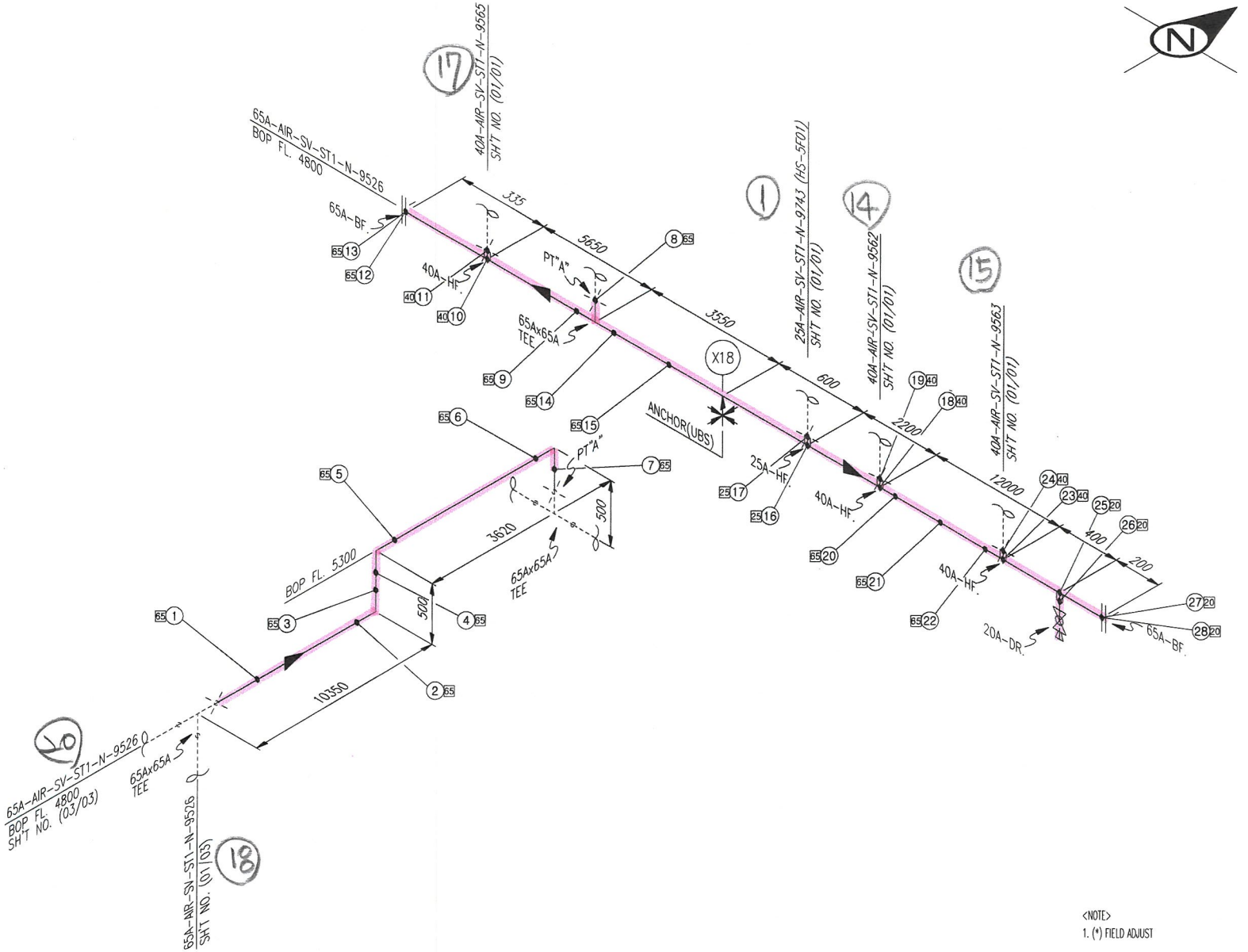
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	-	DESIGN	10	60
			OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050007-C
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 65A-AIR-SV-ST1-N-9526

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 01 / 03
REV. 0



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	65A	39.9M	PIPE, STS 304TP, ERW, BE - SCH 10S
NP1	20A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	65A	3EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
T	65A	1EA	TEE EQUAL, STS304, ERW, BE - SCH 10S
HF	40A	3EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	25A	1EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	20A	1EA	COUPLING HALF, STSF304, SW - SCH 10S
PLG	20A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	65A	2EA	FLG SO, KS 10K, RF, STS304
FB	65A	2EA	FLG BLIND, KS 10K, RF, STS304
G	65A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	65A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

CLIENT SK hynix

19

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

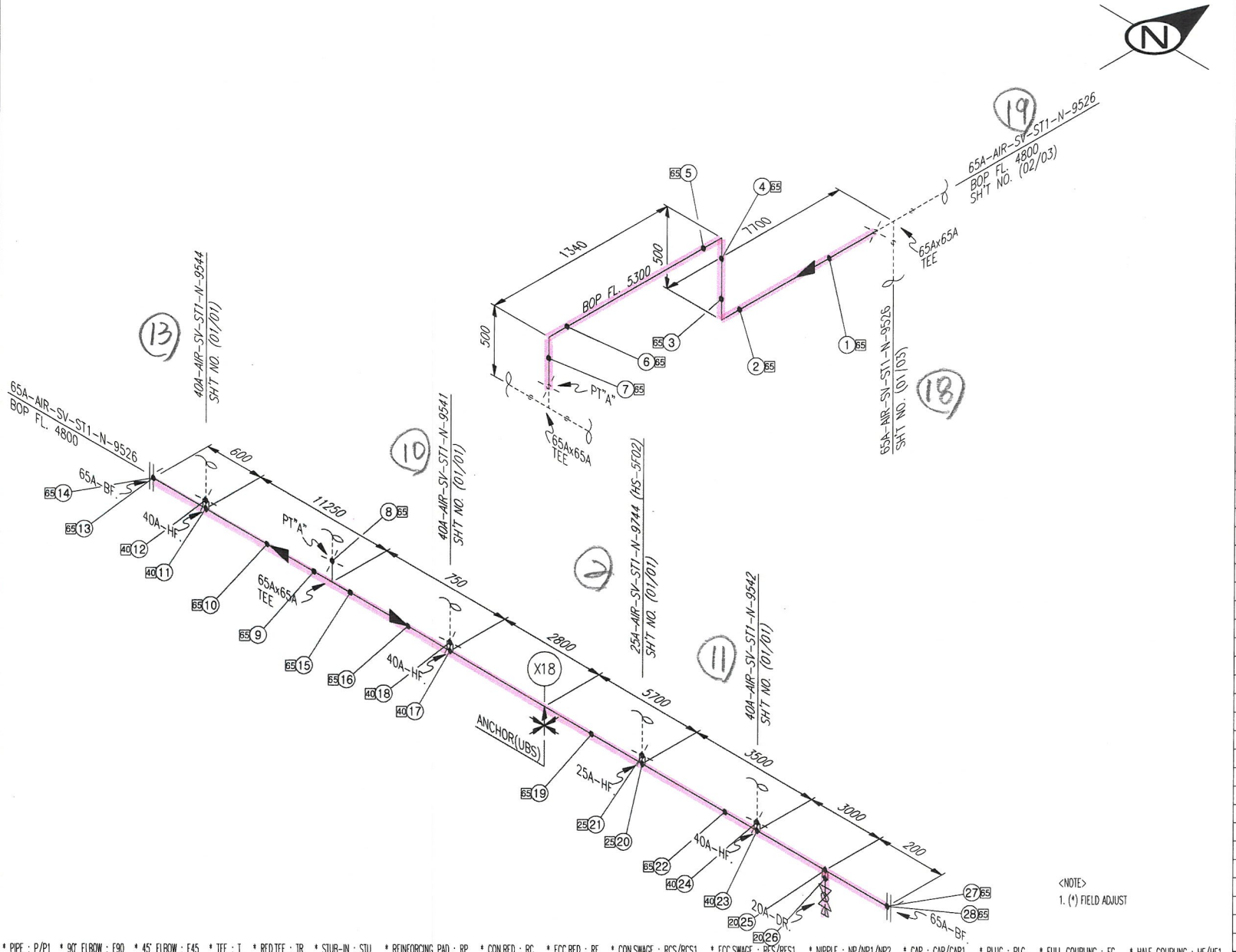
AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050004-D
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 65A-AIR-SV-ST1-N-9526

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 02 / 03
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	65A	37.8M	PIPE, STS 304TP, ERW, BE - SCH 10S
NP1	20A	1EA	NIPPLE, STS 304TP, ERW, TOE - SCH 40S
E90	65A	3EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
T	65A	1EA	TEE EQUAL, STS304, ERW, BE - SCH 10S
HF	40A	3EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	25A	1EA	COUPLING HALF, STSF304, SW - SCH 10S
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PLG	20A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 /
			TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	65A	2EA	FLG SO, KS 10K, RF, STS304
FB	65A	2EA	FLG BLIND, KS 10K, RF, STS304
G	65A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	65A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

CLIENT SK hynix

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N
	THICK.	-

CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
DESIGN	10	60
OPERATING	7.5	AMB

AREA W14
UNIT 5FA
REF. PLAN DWG. CJ-M15W14-PI-A30-050009-D
REF. P & ID CJ-M15W13-PR-A20-000083-A

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영

CONTRACTOR SK ecoplant
COMMODITY AIR-SV
LINE NO. 65A-AIR-SV-ST1-N-9526

PROJECT NAME M15X Ph4 Project
JOB NO. 22330D
SH'T NO. 03 / 03
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

PUNCH LIST

Project No : 22330D

Project Title : M15X Ph3 Project

Package No :

Date :

S/N	Drawing No	Description	Category A B C	Punch Clear Status			
				작성자		확인자	
				Name	Sign	Name	Sign
1							
2							
3							
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12							
13							
14							
15							

Remarks

A : Before Hydro Test

B : Before M/C

C : Before Commissioning